Feedback and Partial Message Side-Information on the Semideterministic Broadcast Channel

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Abstract

The capacity of the semideterministic discrete memoryless broadcast channel (SD-BC) with partial message side-information (P-MSI) at the receivers is established. For the setting without a common message, it is shown that P-MSI to the deterministic receiver can only increase capacity if also the stochastic receiver has P-MSI, while P-MSI to the stochastic receiver alone can increase capacity. For the setting where the encoder also conveys a common message, it is shown that P-MSI to the deterministic receiver only can also increase capacity.

The capacity results are used to show that on the SD-BC with or without P-MSI feedback from the stochastic receiver can increase capacity; with P-MSI at the deterministic receiver it can, in particular, increase the sum-rate capacity. In contrast, as we also show, when the stochastic receiver has full MSI (F-MSI), feedback cannot increase capacity.

The link between feedback and P-MSI is a new feedback code for the SD-BC, which—roughly speaking—turns feedback into P-MSI at the stochastic receiver. The P-MSI allows the stochastic receiver to better mitigate experienced interference.

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1 Introduction

It is well known that feedback can increase the capacity of the discrete memoryless broadcast channel (BC) if both receivers are stochastic [1]. A common intuition for this is that feedback allows the transmitter to create a message that is beneficial to both receivers and, in doing so, to alleviate the interference at each receiver. In previously-presented examples where feedback increases the capacity of the BC, the transmitter typically creates from the feedback a common message comprising information about past noise at the receivers, most prominently in Dueck's example [1].

In this paper, we show that feedback can increase the capacity of the semideterministic BC (SD-BC). Since the SD-BC has one deterministic and one noisy receiver, information about past noise cannot be beneficial to both receivers: It is only beneficial to the stochastic receiver. For the deterministic receiver learning the noise is useless. To prove that feedback can increase capacity on the SD-BC, we thus need a different intuition. Here, we propose the intuition that feedback can increase capacity because it allows the encoder to build up partial message side-information (P-MSI) at the stochastic receiver.

A receiver with P-MSI on the BC is a receiver that has partial knowledge of the message intended for the other receiver. In this paper, we establish the capacity region of the SD-BC when each receiver has P-MSI. Generally, the capacity region with P-MSI at the receivers is larger than without MSI. In particular, P-MSI at the stochastic receiver only can increase capacity on the SD-BC (see Example 1). The symmetric statement is not true: P-MSI at the deterministic receiver only cannot increase capacity (Corollary 6). This holds even when the deterministic receiver has full MSI (F-MSI), i.e., even when it knows the entire message intended for the stochastic receiver. Only when the stochastic receiver has P-MSI, P-MSI at the deterministic receiver can increase capacity (Note 5).

The above results hold for the setting without a common message. No extra effort is required to derive the capacity region of the SD-BC with P-MSI when the encoder also conveys a common message, and we thus establish the capacity region for this more general setting. In contrast to the setting without a common message, P-MSI at the deterministic receiver only can increase capacity if the encoder also conveys a common message (see Example 3).

We use our results for the SD-BC with P-MSI to show that feedback can increase the capacity of the SD-BC with or without P-MSI and, in particular, the sum-rate capacity when the deterministic receiver has P-MSI (Theorem 15). These results are contrasted by our last finding: on the SD-BC feedback cannot increase capacity when the stochastic

receiver has F-MSI (Theorem 16).

To show that feedback increases the capacity of the SD-BC, we introduce a feedback code, which—roughly speaking—turns the feedback into P-MSI at the stochastic receiver. The scheme comprises two transmission phases: Phase 1 and Phase 2. In Phase 1 the encoder codes for an *enhanced BC* that is obtained from the original SD-BC by revealing the outputs observed at the stochastic receiver also to the deterministic receiver. In Phase 2, the sender conveys not only fresh message information but also transmits a compressed version of the stochastic receiver's phase-1 channel outputs to the deterministic receiver, allowing the latter to decode its phase-1 message. Since the stochastic receiver can compute the compressed version of its phase-1 channel outputs, it can use it as P-MSI in Phase 2. At the end of each phase, the stochastic receiver decodes the message sent to it during that phase. The deterministic receiver waits with decoding until the end of Phase 2, when it first decodes its intended phase-2 message and the stochastic receiver's phase-1 channel outputs. It then decodes its intended phase-1 message from its phase-1 channel outputs and its estimate of the stochastic receiver's phase-1 channel outputs.

Based on our feedback code, we can offer the following intuition why feedback increases the capacity of the SD-BC: feedback on the SD-BC (with or without P-MSI) can increase capacity because it allows the encoder to identify a message that is desired at the deterministic receiver and known at the stochastic receiver. In this sense, the feedback can be used to induce a scenario with P-MSI at the stochastic receiver, which allows the stochastic receiver to better mitigate experienced interference. The intuition does not apply when the stochastic receiver has F-MSI. In this case, the stochastic receiver is already able to fully mitigate experienced interference even when there is no feedback. And in fact, feedback does not increase capacity.

To the best of our knowledge, our results are the first to show that feedback can increase the capacity of the SD-BC. While feedback does neither increase the capacity of the deterministic BC nor the capacity of the physically-degraded BC [2], it is known that feedback increases the capacity of several BCs with two stochastic receivers [1, 3, 4, 5]. Achievable regions for the BC with feedback can be found in [3, 4, 5, 6], but even with those at hand showing that feedback can increase the capacity of the SD-BC is nontrivial.

Our feedback code, which uses the feedback to build up MSI at the stochastic receiver, is inspired by the work of Wu and Wigger [5]. They show that for an asymmetric setup where one receiver of the BC is generally stronger than the other, feedback allows

the encoder to identify an extra message for the stronger receiver that describes the weaker receiver's previous channel outputs. Because that extra message is intended for the stronger receiver and known to the weaker receiver, it can be accommodated in the cloud-center of Marton's scheme without causing interference at the weaker receiver. We show that the feedback code of [5] can also improve over the no-feedback capacity of SD-BCs where none of the two receivers is stronger than the other (see Remark 1). This is even more surprising if one recalls that Marton's scheme achieves the capacity of the SD-BC when it is specialized to a satellite-only scheme without cloud-center.

The capacity of the SD-BC without MSI was derived in [7, 8]. Kramer and Shamai [9] introduced and solved the BC with full-MSI (F-MSI) at both receivers, i.e., when each receiver knows the other receiver's message. They also obtained the capacity region of the BC with degraded message sets and P-MSI. P-MSI on the three-receivers BC is discussed in [10]. Independently of our work, Asadi, Ong, and Johnson proposed a coding scheme for general two-receiver BCs with P-MSI [11]. Provided that the auxiliary random variables are chosen appropriately, their scheme coincides with ours (see Remark 2 in Appendix A). It should be noted that this parallel work does not contain a converse for the SD-BC.

Somewhat similar to P-MSI is decoder cooperation on the BC: In this network the decoders have finite-capacity links through which they can provide each other with any information (including information about their messages). The capacity region of the SD-BC with one-sided cooperation via a link from the deterministic to the stochastic receiver is established in [12]. The physically-degraded BC with parallel conferencing decoders and the broadcast channel with degraded message sets are studied in [13].

The remainder of this paper is structured as follows. The subsequent paragraph describes our notation. The second section describes the channel model. The third section is dedicated to the SD-BC with P-MSI, and the fourth section to the SD-BC with feedback, with and without P-MSI at the receivers.

1.1 Notation and Preliminaries

For two integers a and b satisfying $a \leq b$ we denote by $[a : b]$ the set $\{a, a+1, \ldots, b\}$. We use calligraphic letters to denote finite sets and $|\cdot|$ for their cardinality, e.g., $\mathcal{X}$ is the input alphabet of a channel and $|\mathcal{X}|$ its cardinality. By $\mathscr{C}$ we refer to a capacity region and by $\partial\mathscr{C}$ to its boundary.

Random variables are denoted by upper-case letters and their realizations or the elements of their support sets by lower-case letters, e.g., Z denotes the random channel

output at the stochastic receiver and $z \in \mathcal{Z}$ a value it may take.

By $Q \sim \text{Unif}[1:n]$ we indicate that the random variable Q is uniformly drawn from the set $[1:n]$. We write $S \sim \text{Ber}(p)$, where $p \in [0,1]$, to indicate that S is a binary random variable that assumes the value 1 with probability p. By $h_{\text{b}}(\cdot)$ we denote the binary entropy function and by $h_{\text{b}}^{-1}(\cdot)$ its inverse on $[0,1/2]$, i.e., $h_{\text{b}}(p)$ is the entropy of the binary random variable S, and for $p \in [0,1/2]$ we have $h_{\text{b}}^{-1}\big(h_{\text{b}}(p)\big) = p$.

Codewords and information sent over a rate-limited feedback link are denoted by bold face lower-case or upper-case letters depending on whether they are deterministic or random, e.g., $\mathbf{x}(m_{\mathcal{Y}}, m_{\mathcal{Z}})$ is the codeword corresponding to the message-pair $(m_{\mathcal{Y}}, m_{\mathcal{Z}})$. The integer n stands for the block-length, and unless otherwise specified sequences are assumed to be of length n.

We shall often say that a set is of size 2^{nR}, where R is some positive real number, without verifying that 2^{nR} is an integer. More accurately, one should write $\lfloor 2^{nR} \rfloor$. In the regime of interest where $n \to \infty$, the difference between the two quantities becomes, however, negligible, and we therefore ignore this technicality. Similarly, we shall assume that a set of size 2^{nR} can be written as the product of any two sets of size 2^{nR_a} and 2^{nR_b}, respectively, whenever R_a and R_b are nonnegative real numbers satisfying $R_a + R_b = R$.

Variables that occur at time i are characterized by a subscript i, e.g., X_i denotes the time-i channel input. Sequences of variables that occur in the time range j to i bear a subscript j and a superscript i, where the subscript $j = 1$ may be dropped, e.g., X_4^5 denotes the forth and fifth channel input, and X^n denotes the entire channel input sequence.

A joint probability mass function (PMF), its marginal PMF, and its conditional PMF are all denoted by the same function $p(\cdot)$, with the exact meaning specified by the subscripts or arguments, e.g., $p_{X,Y}(0,1)$ denotes the probability of the event $(X,Y) = (0,1)$ and $p(x|y)$ the probability that $X = x$ given $Y = y$.

We denote ϵ-typical sequences of length n by $\mathcal{T}_\epsilon^{(n)}$, e.g., $\mathcal{T}_\epsilon^{(n)}(X,Y)$ denotes the set of length-n sequences $(x^n, y^n) \in \mathcal{X}^n \times \mathcal{Y}^n$ that are jointly ϵ-typical w.r.t. $p_{X,Y}(\cdot,\cdot)$ (see [14, Chapter 2]). By $\delta(\cdot)$ we denote a function of ϵ that converges to 0 as ϵ approaches 0. Moreover, we denote by $\{\epsilon_n\}$ any sequence of numbers that converges to 0 as n tends to infinity.

We shall use the functional representation lemma:

Lemma 1. *[15, Functional Representation Lemma] Given two discrete random variables X and Y, there exists a discrete random variable S, which is independent of X, and a function $g\colon \mathcal{X} \times \mathcal{S} \to \mathcal{Y}$ such that $Y = g(X,S)$.*

2 Channel Model

The SD-BC with finite input alphabet $\mathcal{X}$ and finite output alphabets $\mathcal{Y}$ and $\mathcal{Z}$ is characterized by a transition law of the form

$$W(y, z|x) = \mathbb{1}_{\{y=f(x)\}} W(z|x).$$

The encoder transmits an n-tuple X^n in order to convey the message-pairs $(M, M_{\mathcal{Y}})$ and $(M, M_{\mathcal{Z}})$ to the deterministic receiver $\mathcal{Y}$ and the stochastic receiver $\mathcal{Z}$, respectively, where M denotes the common message and $M_{\mathcal{Y}}$ and $M_{\mathcal{Z}}$ the private messages. We assume that M, $M_{\mathcal{Y}}$, and $M_{\mathcal{Z}}$ are independent, that M us uniformly drawn from a set of size 2^{nR}, and that for each $\nu \in \{\mathcal{Y}, \mathcal{Z}\}$ message M_ν is uniformly drawn from a set of size 2^{nR_ν}. We study the SD-BC with P-MSI, and we thus assume that each private message comprises two parts, i.e.,

$$M_\nu = M_\nu^{(p)} \times M_\nu^{(c)},\ \nu \in \{\mathcal{Y}, \mathcal{Z}\},$$

that receiver $\mathcal{Y}$ knows $M_{\mathcal{Z}}^{(c)}$ and decodes $(M, M_{\mathcal{Y}})$ from $(Y^n, M_{\mathcal{Z}}^{(c)})$, and that receiver $\mathcal{Z}$ knows $M_{\mathcal{Y}}^{(c)}$ and decodes $(M, M_{\mathcal{Z}})$ from $(Z^n, M_{\mathcal{Y}}^{(c)})$. For each $\nu \in \{\mathcal{Y}, \mathcal{Z}\}$ we assume that $M_\nu^{(p)}$ and $M_\nu^{(c)}$ are independent of each other and uniformly drawn from sets of size $2^{nR_\nu^{(p)}}$ and $2^{nR_\nu^{(c)}}$, respectively, where $R_\nu = R_\nu^{(p)} + R_\nu^{(c)}$. Note the extreme cases:

- $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}) = (R_{\mathcal{Y}}, 0) \quad \Longleftrightarrow \quad$ no MSI at receiver $\mathcal{Z}$
- $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}) = (0, R_{\mathcal{Y}}) \quad \Longleftrightarrow \quad$ F-MSI at receiver $\mathcal{Z}$
- $(R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) = (R_{\mathcal{Z}}, 0) \quad \Longleftrightarrow \quad$ no MSI at receiver $\mathcal{Y}$
- $(R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) = (0, R_{\mathcal{Z}}) \quad \Longleftrightarrow \quad$ F-MSI at receiver $\mathcal{Y}$.

A rate-tuple $(R, R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ is called achievable if there exists a sequence of encoders and decoders such that at each receiver the probability of a decoding error tends to zero as n tends to infinity. The capacity region is the closure of the set of all achievable rate-tuples.

We study the SD-BC with P-MSI in the absence and in the presence of feedback. In the absence of feedback, the encoder selects the channel input sequence as a function of the triple $(M, M_{\mathcal{Y}}, M_{\mathcal{Z}})$, i.e., $X^n = \phi(M, M_{\mathcal{Y}}, M_{\mathcal{Z}})$. This setting corresponds to that of Figure 1 without the dashed links.

When there is feedback, it is assumed to be one-sided from the stochastic receiver $\mathcal{Z}$ only. (Feedback from the deterministic receiver $\mathcal{Y}$ is useless because the transmitter

can always compute Y^n from X^n.) We consider both perfect and rate-limited feedback. Perfect feedback allows the encoder to form the time-i input also as a function of Z^{i-1}, i.e., $X_i = \phi_i(M, M_{\mathcal{Y}}, M_{\mathcal{Z}}, Z^{i-1})$, $i \in [1:n]$. Rate-limited feedback of rate R_{FB} allows receiver $\mathcal{Z}$ to transmit after Transmission i a signal $W_i = h(Z^i, M_{\mathcal{Y}}^{(c)}) \in \mathcal{W}_i$ to the encoder, and the encoder can in turn form the time-i input also as a function of W^{i-1}, i.e., $X_i = \phi_i(M, M_{\mathcal{Y}}, M_{\mathcal{Z}}, W^{i-1})$, $i \in [1:n]$. The rate-limitation implies that $\prod_{i=1}^n |\mathcal{W}_i| \leq 2^{nR_{\text{FB}}}$. The SD-BC with P-MSI and perfect feedback (rate-limited feedback) corresponds to the setting of Figure 1 where the dashed links transport the chance variable Z_i (W_i). Note that perfect feedback is more powerful than rate-limited feedback: any rate-tuple that is achievable with rate-limited feedback can also be achieved with perfect feedback. Rate-limited and perfect feedback are equally powerful when $R_{\text{FB}} \geq \log|\mathcal{Z}|$.

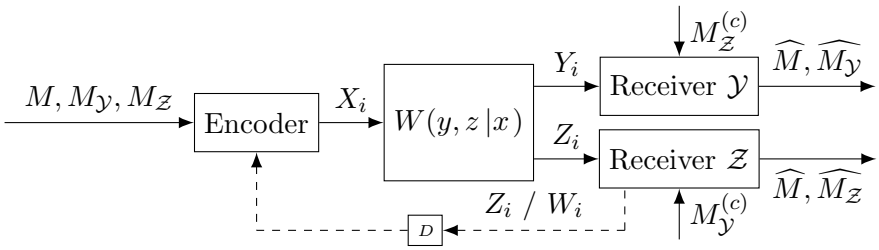

Figure 1: SD-BC with P-MSI and feedback.

3 The SD-BC with P-MSI

In the absence of feedback, our main result is a single-letter characterization of the capacity of the SD-BC with P-MSI:

Theorem 2 (Capacity with P-MSI)**.** *The capacity region $\mathscr{C}_{\text{P-MSI}}$ of the SD-BC with P-MSI is the set of rate-tuples $(R, R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying*

$$R + R_{\mathcal{Y}} \leq H(Y) \tag{1a}$$

$$R + R_{\mathcal{Z}} \leq I(U;Z) \tag{1b}$$

$$R + R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq I(V;Y) + H(Y|U) + I(U;Z|V) \tag{1c}$$

$$R + R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}} \leq H(Y|U) + I(U;Z) \tag{1d}$$

$$2R + R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq I(V;Y) + H(Y|U) + I(U;Z) \tag{1e}$$

for some PMF of the form

$$p(v,u,x,y,z) = p(v,u)p(x|u)W(y,z|x). \tag{2}$$

W.l.g., one can restrict X to be a function of (Y,U).

Proof. See Appendix A. □

In the following, we sketch and discuss the proof of the direct part. The capacity-achieving code is rigorously described in Appendix A. Here, we try to provide some intuition. A tentative code construction is as follows. Use Marton's code construction (see [14, Section 8.4]) that encodes the common message tuple $(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$ into a cloud-center V^n and the private messages $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$ into satellites Y^n and U^n. Receiver $\mathcal{Y}$ decodes $(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$ and $M_{\mathcal{Y}}^{(p)}$ jointly, while taking into account its knowledge of $M_{\mathcal{Z}}^{(c)}$, and likewise receiver $\mathcal{Z}$ decodes $(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$ and $M_{\mathcal{Z}}^{(p)}$ jointly, while taking into account its knowledge of $M_{\mathcal{Y}}^{(c)}$.

The above code can achieve every rate-tuple $(R, R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ that satisfies (1) and

$$R_{\mathcal{Y}}^{(p)} \leq H(Y|V) \tag{3a}$$

$$R_{\mathcal{Z}}^{(p)} \leq I(U;Z|V) \tag{3b}$$

$$R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}}^{(p)} \leq H(Y|U) + I(U;Z|V) \tag{3c}$$

for some PMF of the form (2). This region differs from the capacity region in Theorem 2 in that the rates $R_{\mathcal{Y}}^{(p)}$ and $R_{\mathcal{Z}}^{(p)}$ must also satisfy (3). As we shall see, it is, in general, strictly smaller than the capacity region (see Example 2 ahead).

To get rid of the constraints (3) and thus achieve the entire capacity region, a fix is needed: the encoder must be able to convey more information about $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$ by allowing the cloud-center V^n to depend not only on the triple $(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$ but also on $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$. To this end, we use the following code construction, which is depicted in Figure 3 for the setting without a common message. (In Figure 3, each dot represents an n-length codeword.) We generate for each triple $(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)})$ a bin containing $2^{n\tilde{R}^{(c)}}$ n-tuples $\mathbf{v}$, which are drawn independently of each other and each from the PMF $\prod_{i=1}^{n} p(v_i)$. (In Figure 3, the light blue row indicates all bins and codewords that are associated with message $M_{\mathcal{Y}}^{(c)} = 2^{nR_{\mathcal{Y}}^{(c)}} - 1$, and the light green column indicates all bins and codewords that are associated with message $M_{\mathcal{Z}}^{(c)} = 2^{nR_{\mathcal{Z}}^{(c)}} - 1$.) For each bin of the cloud-center, we generate two satellite codebooks: one to encode $M_{\mathcal{Y}}^{(p)}$ and one to encode $M_{\mathcal{Z}}^{(p)}$. (Figure 3 shows the satellite codebooks corresponding

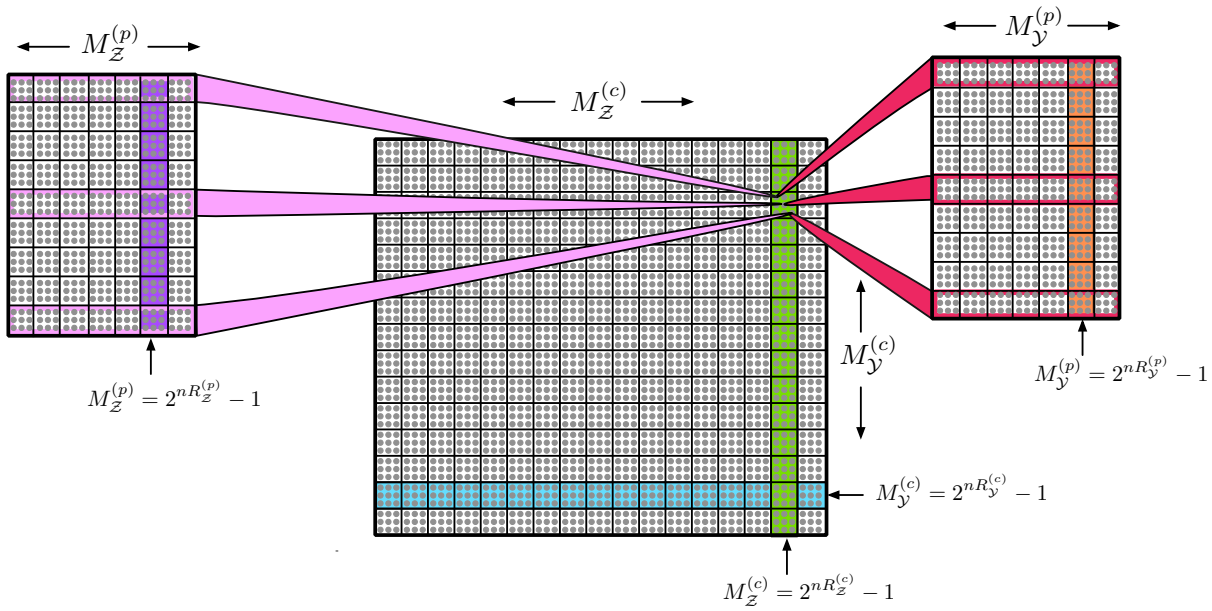

Figure 2: Code construction without common message.

to $(M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}) = (2^{nR_{\mathcal{Y}}^{(c)}} - 1, 2^{nR_{\mathcal{Z}}^{(c)}} - 1)$. The satellite codebook for $M_{\mathcal{Y}}^{(p)}$ is on the right and the satellite codebook for $M_{\mathcal{Z}}^{(p)}$ on the left.) The first $2^{n(\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)})}$ $\mathbf{y}$-codewords in each satellite codebook for $M_{\mathcal{Y}}^{(p)}$ are superpositioned on the first codeword in the corresponding cloud-center bin; the following $2^{n(\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)})}$ codewords in each satellite codebook are superpositioned on the second codeword in the corresponding cloud-center bin; and so on. That is, the first $2^{n(\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)})}$ $\mathbf{y}$-codewords are drawn according to the conditional PMF $\prod_{i=1}^{n} p(y_i|v_i^{(1)})$, where $v_i^{(1)}$ denotes the i-th component of the first codeword in the corresponding clound-center bin; the following $2^{n(\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)})}$ $\mathbf{y}$-codewords are drawn according to the conditional PMF $\prod_{i=1}^{n} p(y_i|v_i^{(2)})$, where $v_i^{(2)}$ denotes the i-th component of the second codeword in the corresponding clound-center bin; and so on. The codewords in the satellite codebooks for $M_{\mathcal{Z}}^{(p)}$ are drawn similarly. (In Figure 3, the codewords in the first three rows of the two satellite codebooks are superpositioned on the first codeword of the cloud-center bin corresponding to $(M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}) = (2^{nR_{\mathcal{Y}}^{(c)}} - 1, 2^{nR_{\mathcal{Z}}^{(c)}} - 1)$.) The codewords in each satellite codebook are again partitioned into bins, where each bin is associated with one specific realization of the message $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$, respectively. (In Figure 3, the bin in orange is associated with message $M_{\mathcal{Y}}^{(p)} = 2^{nR_{\mathcal{Y}}^{(p)}} - 1$ and the bin in purple is associated with message $M_{\mathcal{Z}}^{(p)} = 2^{nR_{\mathcal{Z}}^{(p)}} - 1$.)

To transmit the message-tuple $(m, m_{\mathcal{Y}}^{(p)}, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(p)}, m_{\mathcal{Z}}^{(c)})$, the encoder looks for n-

tuples $\mathbf{v}$, $\mathbf{y}$, and $\mathbf{u}$ in the bins corresponding to $(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)})$, $m_{\mathcal{Y}}^{(p)}$, and $m_{\mathcal{Z}}^{(p)}$, respectively, such that $(\mathbf{v}, \mathbf{y}, \mathbf{u})$ are jointly typical. It then generates the channel input X^n from the product distribution $\prod_{i=1}^{n} p(x_i|y_i, u_i)$.[1] Receiver $\mathcal{Y}$ decodes $(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$ and $M_{\mathcal{Y}}^{(p)}$ jointly, while restricting attention to the column of the cloud-center that corresponds to the message $M_{\mathcal{Z}}^{(c)}$, which receiver $\mathcal{Y}$ knows. Likewise, receiver $\mathcal{Z}$ decodes $(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$ and $M_{\mathcal{Z}}^{(p)}$ jointly, while restricting attention to the row of the cloud-center that corresponds to the message $M_{\mathcal{Y}}^{(c)}$, which receiver $\mathcal{Z}$ knows.

As we explain in Appendix A (see Remark 2 ahead), the effect of binning the cloud-center is the same as that of rate-splitting, where the messages $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$ are divided into two parts $M_{\mathcal{Y}}^{(p)} = (M_{\mathcal{Y},s}^{(p)}, M_{\mathcal{Y},c}^{(p)})$ and $M_{\mathcal{Z}}^{(p)} = (M_{\mathcal{Z},s}^{(p)}, M_{\mathcal{Z},c}^{(p)})$, and where only $M_{\mathcal{Y},s}^{(p)}$ and $M_{\mathcal{Z},s}^{(p)}$ are associated with the satellites Y^n and U^n, respectively, whereas $M_{\mathcal{Y},c}^{(p)}$ and $M_{\mathcal{Z},c}^{(p)}$ are encoded in the cloud-center. The benefit of binning the cloud-center is that it only necessitates one auxiliary rate: the rate at which the cloud-center is binned. In contrast, rate-splitting necessitates two auxiliary rates: the rates $R_{\mathcal{Y},c}^{(p)}$ and $R_{\mathcal{Z},c}^{(p)}$ associated with $M_{\mathcal{Y},c}^{(p)}$ and $M_{\mathcal{Z},c}^{(p)}$, respectively.

We next discuss several special cases where one or both of the receivers have no MSI or F-MSI. Section 3.1 contains the results for the setting without a common message, and Section 3.2 is concerned with the setting where the encoder also conveys a common message. Based on the results, we show that without a common message P-MSI to the deterministic receiver can only increase capacity if also the stochastic receiver has P-MSI, while P-MSI to the stochastic receiver alone can increase capacity. The situation is different when the encoder also conveys a common message; with a common message P-MSI to the deterministic receiver alone can also increase capacity.

3.1 Without a Common Message

This section considers the setting without a common message, i.e., when $R = 0$. Without any MSI, i.e., when $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$ and $R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$, Constraints (1c) and (1e) are redundant, and we recover:

Corollary 3 (No MSI)**.** *[From [8, 7], see also Section 8.3.1 of [14]] The capacity region $\mathscr{C}$ of the SD-BC without MSI is the set of rate-triples $(R, R_{\mathcal{Y}}, R_{\mathcal{Z}})$ satisfying*

$$R_{\mathcal{Y}} \leq H(Y) \tag{4a}$$

$$R_{\mathcal{Z}} \leq I(U; Z) \tag{4b}$$

[1] It is well-known that without a cloud-center, this scheme achieves the capacity of the SD-BC without a common message and without MSI (see e.g. [14, Sections 8.3.1–8.3.2]).

$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq H(Y|U) + I(U;Z) \tag{4c}$$

for some PMF of the form

$$p(u,x,y,z) = p(u,x)W(y,z|x). \tag{5}$$

W.l.g., one can restrict X to be a function of (Y,U).

Proof. If we choose V to be deterministic, then (1) and (4) are equivalent. Hence, (4) is an inner bound. Moreover, (1a), (1b), and (1d) imply that this inner bound is tight. □

With F-MSI at the stochastic receiver $\mathcal{Z}$, i.e., when $R_{\mathcal{Y}}^{(p)} = 0$, Theorem 2 gives:

Corollary 4 (F-MSI at $\mathcal{Z}$)**.** *For each $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$, the capacity region of the SD-BC with F-MSI at receiver $\mathcal{Z}$ ($R_{\mathcal{Y}}^{(p)} = 0$) is the set of rate-tuples $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying*

$$R_{\mathcal{Y}} \leq H(Y) \tag{6a}$$

$$R_{\mathcal{Z}} \leq I(X;Z) \tag{6b}$$

$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq I(X;Y,Z) \tag{6c}$$

for some PMF of the form

$$p(x,y,z) = p(x)W(y,z|x). \tag{7}$$

Proof. See Appendix B. □

The corollary can be used to show the following result:

Note 5 (P-MSI Helps)**.** *On the SD-BC, P-MSI to the stochastic receiver only can increase capacity, and when the stochastic receiver has P-MSI, then P-MSI to the deterministic receiver can also increase capacity.*

We prove the note by means of an example:

Example 1 (P-MSI Helps)**.** *Consider the SD-BC with binary input X, state $S \sim \mathrm{Ber}(p)$, $p \in (0, 1/2)$, and outputs $Y = X$ and $Z = X \oplus S$. As shown in [14, Section 5.4.2], the capacity region $\mathscr{C}$ without MSI is the set of rate-pairs $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ that satisfy for some $\alpha \in [0, 1/2]$*

$$R_{\mathcal{Y}} \leq h_b(\alpha) \tag{8a}$$

$$R_{\mathcal{Z}} \leq 1 - h_b(\alpha + p - 2\alpha p). \tag{8b}$$

In particular, the sum-rate is bounded away from 1 *unless* $R_{\mathcal{Z}} = 0$. *By Corollary 4 (with* $X \sim \mathrm{Ber}(1/2)$*), the capacity region* $\mathscr{C}_{\text{P-MSI}}$ *with F-MSI at the stochastic receiver* $\mathcal{Z}$ *(*$R_{\mathcal{Y}}^{(p)} = 0$*) is the set of rate-tuples* $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ *satisfying*

$$R_{\mathcal{Y}} \leq 1 \tag{9a}$$

$$R_{\mathcal{Z}} \leq 1 - h_b(p) \tag{9b}$$

$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq 1. \tag{9c}$$

When the deterministic receiver $\mathcal{Y}$ *has no MSI (*$R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$*) and the stochastic receiver* $\mathcal{Z}$ *has F-MSI (*$R_{\mathcal{Y}}^{(p)} = 0$*), the sum-rate can thus be* 1 *even when* $R_{\mathcal{Z}} > 0$. *When both receivers have F-MSI (*$R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Y}}^{(p)} = 0$*), then the sum-rate can be larger than* 1.

In contrast, P-MSI at the deterministic receiver $\mathcal{Y}$ only cannot increase capacity:

Corollary 6 (No MSI at $\mathcal{Z}$)**.** *When the stochastic receiver* $\mathcal{Z}$ *has no MSI (*$R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$*), then irrespective of* $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$

$$(R_{\mathcal{Y}}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}} \quad \Longleftrightarrow \quad (R_{\mathcal{Y}}, R_{\mathcal{Z}}) \in \mathscr{C}, \tag{10}$$

i.e., the capacity region of the SD-BC with P-MSI at the deterministic receiver is the set of rate-tuples $(R_{\mathcal{Y}}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ *satisfying*

$$R_{\mathcal{Y}} \leq H(Y) \tag{11a}$$

$$R_{\mathcal{Z}} \leq I(U; Z) \tag{11b}$$

$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq H(Y|U) + I(U; Z) \tag{11c}$$

for some PMF of the form

$$p(u, x, y, z) = p(u, x) W(y, z|x). \tag{12}$$

W.l.g., one can restrict X *to be a function of* (Y, U).

Proof. The direct part is due to Corollary 3, and the converse follows from (1a), (1b), and (1d). □

Hinging on the above capacity results, we next show that for the setting without a common message the rate region that is achievable without binning the cloud-center (or, equivalently, without rate-splitting) is in general strictly smaller than the capacity region of the SD-BC with P-MSI. Recall that if the coud-center is not binned, then we can achieve every rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ that satisfies (1) and (3) for some PMF of the form (2). Moreover, by the functional representation lemma (Lemma 1)

we can w.l.g. restrict X to be a function of (Y, U) (this can be shown similarly as in Appendix A.2). Note that the rate-region that is achievable without binning the cloud-center is contained in the capacity region $\mathscr{C}_{\text{P-MSI}}$ of the SD-BC with P-MSI in Theorem 2. The following example shows that the containment can be strict.

Example 2 (Binning of Cloud-Center Needed)**.** *Consider the SD-BC with binary input X, state $S \sim \text{Ber}(p)$, $p \in [0,1]$, and outputs $Y = X$ and*

$$Z = \begin{cases} X & \text{if } S = 0, \\ ? & \text{if } S = 1. \end{cases}$$

On account of Corollary 4, the capacity region $\mathscr{C}_{\text{P-MSI}}$ with F-MSI at the stochastic receiver $\mathcal{Z}$ ($R_{\mathcal{Y}}^{(p)} = 0$) is the set of rate-tuples $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying

$$R_{\mathcal{Y}} \leq 1 \tag{13}$$

$$R_{\mathcal{Z}} \leq 1 - p \tag{14}$$

$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq 1. \tag{15}$$

In particular, it contains the rate-tuple $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) = (0, p, 1-p, 0)$ for any given $p \in [0,1]$. As it is shown in Appendix C, if we do not bin the cloud-center, then the rate-tuple $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) = (0, p, 1-p, 0)$ is achievable only for $p \in \{0, 1\}$. Hence, without binning the cloud-center we can only achieve a strict subset of the capacity region.

3.2 With a Common Message

This section considers the setting with a common message, i.e., when $R > 0$. With F-MSI at the stochastic receiver $\mathcal{Z}$, i.e., when $R_{\mathcal{Y}}^{(p)} = 0$, Theorem 2 gives:

Corollary 7 (F-MSI at $\mathcal{Z}$)**.** *For each $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$, the capacity region of the SD-BC with F-MSI at receiver $\mathcal{Z}$ ($R_{\mathcal{Y}}^{(p)} = 0$) is the set of rate-tuples $(R, 0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying*

$$R + R_{\mathcal{Y}} \leq H(Y) \tag{16a}$$

$$R + R_{\mathcal{Z}} \leq I(X; Z) \tag{16b}$$

$$R + R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq I(X; Y, Z) \tag{16c}$$

for some PMF of the form

$$p(x, y, z) = p(x) W(y, z|x). \tag{17}$$

Proof. For $U = X$ and $V = Y$ the constraints in (1) and (16) are equivalent. Hence, (16) is an inner bound on the capacity region. We next argue that (16) is also an outer bound on the capacity region. To this end, fix any PMF of the form (2) satisfying that X is a function of (Y, U). It is clear that

$$I(U; Z) \leq I(X; Z), \tag{18}$$

and from the proof of Corollary 4 (see Equation (88)) we know that

$$I(V; Y) + H(Y|U) + I(U; Z|V) \leq I(X; Y, Z). \tag{19}$$

From (1a), (1b) and (18), as well as (1c) and (19) we conclude that (16) is also an outer bound on the capacity region. □

With F-MSI at the deterministic receiver $\mathcal{Y}$, i.e., when $R_{\mathcal{Z}}^{(p)} = 0$, Theorem 2 gives:

Corollary 8 (F-MSI at $\mathcal{Y}$)**.** *For each $R_{\mathcal{Y}}^{(p)} \in [0, R_{\mathcal{Y}}]$, the capacity region of the SD-BC with F-MSI at receiver $\mathcal{Y}$ ($R_{\mathcal{Z}}^{(p)} = 0$) is the set of rate-tuples $(R, R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, 0, R_{\mathcal{Z}})$ satisfying*

$$R + R_{\mathcal{Y}} \leq H(Y) \tag{20a}$$

$$R + R_{\mathcal{Z}} \leq I(U; Z) \tag{20b}$$

$$R + R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}} \leq H(Y|U) + I(U; Z) \tag{20c}$$

for some PMF of the form

$$p(u, x, y, z) = p(u, x)W(y, z|x). \tag{21}$$

Proof. For $V = U$ the constraints in (1) and (20) are equivalent. Hence, (20) is an inner bound on the capacity region. That (20) is also an outer bound on the capacity region follows from (1a), (1b), and (1d). □

In contrast to the setting without a common message, where MSI to the deterministic receiver only cannot increase capacity (see Corollary 6), it follows from Corollary 8 that in the setting with a common message MSI to the deterministic receiver only can increase capacity:

Note 9 (P-MSI at $\mathcal{Y}$ Helps)**.** *On the SD-BC with a common message, P-MSI to the deterministic receiver only can increase capacity.*

We prove the note by means of an example:

Example 3 (P-MSI at $\mathcal{Y}$ Helps)**.** *Consider the SD-BC with binary inputs X_1 and X_2, binary state $S \sim \mathrm{Ber}(p)$, $p \in (0,1)$, and outputs $Y = X_1 + X_2$ and*

$$Z = \begin{cases} X_2 & S = 0, \\ ? & S = 1. \end{cases}$$

If the deterministic receiver $\mathcal{Y}$ has F-MSI ($R_{\mathcal{Z}}^{(p)} = 0$) and the stochastic receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$), then it follows from (20c) *and the proof of Example 4 ahead (see Equations* (133)–(135)*) that the maximum achievable sum-rate is*

$$R + R_{\mathcal{Y}} + R_{\mathcal{Z}} = H(Y|U) + I(U;Z) = \bar{p} + \left[p h_b\left(\frac{1}{1+2^{1/p}} \right) + \frac{2^{1/p}}{1+2^{1/p}} \right], \tag{22}$$

where every PMF $p(u,x,y,z)$ of the form (21) *that achieves the maximum sum-rate satisfies*

$$I(U;Y) < \min\{H(Y), I(U;Z)\}. \tag{23a}$$

Moreover, (20) *readily implies that the maximum sum-rate* (22) *is achievable for every*

$$R \in \big[0, \min\{H(Y), I(U;Z)\}\big]. \tag{24}$$

As it is shown in Appendix D, in the absence of MSI, i.e., when $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$ and $R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$, we can achieve the maximum sum-rate (22) *only if*

$$R \leq I(V;Y) \leq I(U;Y). \tag{25}$$

Because of (23a)*, this implies that if the encoder also conveys a common message, then P-MSI to the deterministic receiver only can increase capacity.*

3.3 Intuition on the Results

In this paragraph, we first provide intuition on why in the setting without a common message P-MSI at the stochastic receiver only can increase capacity, but P-MSI at the deterministic receiver only cannot increase capacity. Then, we explain why P-MSI at the deterministic receiver can increase capacity if either the stochastic receiver has P-MSI or if the P-MSI is available only at the deterministic receiver but the encoder also conveys a common message to both receivers.

On the two-receiver BC with independent messages, the transmission to each of the two receivers is interfered by the transmission to the other receiver. Since the transmitter

knows the two messages and thus (non-causally) the two transmissions, it can partially mitigate the interference experienced at each receiver (see Marton's scheme [7]).

When one of the two receivers has P-MSI, it has knowledge about the transmission to the other receiver. In general, knowing interference both at the transmitter and the receiver is more beneficial (in terms of achievable rates) than knowing it only at the transmitter (cf. the single-user channel with non-causal state information). As a consequence, with P-MSI a receiver can better mitigate the interference caused to its intended transmission than without MSI, and thus P-MSI at only one receiver typically increases capacity. The situation is different when one of the two receivers has a deterministic channel. In fact, if the channel output of a single-user channel is a function of the channel input and some interference, e.g. a state sequence, then knowing interference non-causally only at the transmitter is as beneficial as knowing it also at the receiver. This may provide intuition on why in the setting without a common message P-MSI at the deterministic receiver only cannot increase capacity, but P-MSI at the stochastic receiver only can.

Interference can be mitigated more efficiently when both receivers have P-MSI. For example, if $R_{\mathcal{Y}}^{(c)} = R_{\mathcal{Z}}^{(c)}$, then the transmitter can send the 'x-or' of the messages $M_{\mathcal{Y}}^{(c)}$ and $M_{\mathcal{Z}}^{(c)}$ as a common message to both receivers.

For the setting with a common message, we have seen that—in contrast to the setting without a common message—P-MSI can also increase capacity if it is only available at the deterministic receiver. A reason might be that the encoder conveys the common message in the cloud-center of Marton's inner bound, and that the cloud-center could be easier to decode for the stochastic receiver than for the deterministic receiver. In such a scenario, one can add an extra message for the stochastic receiver to the cloud-center. If the deterministic receiver already knows this extra message, e.g., because it has P-MSI, then this is possible without making the decoding at the deterministic receiver unreliable.

4 Feedback on the SD-BC without Common Message

This section investigates how feedback can be used to further mitigate interference on the SD-BC without a common message. We consider the cases with and without P-MSI at the receivers. Since feedback from the deterministic receiver $\mathcal{Y}$ is useless, we consider perfect or rate-limited feedback from the stochastic receiver $\mathcal{Z}$ only.

In the absence of MSI, (even one-sided) feedback can increase the capacity of the BC if both receivers are stochastic [1]. A common intuition for this is that feedback

allows the transmitter to create a message that is beneficial to both receivers and, in doing so, to alleviate the interference at each receiver. In previously-presented examples, the transmitter typically creates a common message comprising information about past noise at the receivers, most prominently in Dueck's example [1].

The SD-BC has only one noisy receiver, and for the deterministic receiver learning the noise is useless. Thus, to prove that feedback can increase capacity on the SD-BC, a different intuition is needed. This section presents a feedback coding scheme, which can increase capacity on the SD-BC with and without P-MSI (see Theorem 15). The scheme allows the encoder to build up P-MSI at the stochastic receiver. The motivation for building up P-MSI at the stochastic receiver is that with P-MSI the stochastic receiver can better mitigate experienced interference (Note 5).

4.1 Coding Scheme with Rate-Limited Feedback

For simplicity of exposition, we assume that only the deterministic receiver $\mathcal{Y}$ has P-MSI but not the stochastic receiver $\mathcal{Z}$, i.e., $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$. We fix $\alpha \in (0,1)$. Then, we partition the blocklength-n transmission into two phases: Phase 1 of length αn and Phase 2 of length $(1-\alpha)n$. In Phase 1 the encoder codes for an *enhanced BC* (with P-MSI at the deterministic receiver $\mathcal{Y}$) at rates $(\tilde{R}_{\mathcal{Y}}^{(p)}, 0, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}^{(\text{enh})}$. The enhanced BC is obtained from the original SD-BC by revealing the outputs observed at the stochastic receiver $\mathcal{Z}$ also to the deterministic receiver $\mathcal{Y}$. Its capacity region $\mathscr{C}_{\text{P-MSI}}^{(\text{enh})}$ is characterized in Proposition 10 ahead. As we explain next, we use the feedback to convey receiver $\mathcal{Z}$'s phase-1 channel outputs to receiver $\mathcal{Y}$, allowing the latter to decode its phase-1 message. After Phase 1, the stochastic receiver $\mathcal{Z}$ compresses its phase-1 channel outputs and sends the compression index over the rate-limited feedback-link to the transmitter, which then conveys it to the deterministic receiver $\mathcal{Y}$ at rate $R_{\mathcal{Y}}^{(c)}$ in the subsequent Phase 2.[2] Simultaneously, the transmitter sends fresh message information at rates $(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ to the two receivers. The stochastic receiver $\mathcal{Z}$ knows the compression index and can thus use it as P-MSI in Phase 2. Hence, in Phase 2 the encoder codes for the SD-BC with P-MSI at *both* receivers, and to guarantee a small phase-2 decoding error probability it is thus enough to have $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}$. Note that only receiver $\mathcal{Y}$ needs to be able to reconstruct receiver $\mathcal{Z}$'s channel outputs $Z^{\alpha n}$. Since receiver $\mathcal{Y}$ can use its own phase-1 outputs $Y^{\alpha n}$ as side-information,

[2] The stochastic receiver $\mathcal{Z}$ uses the feedback-link only between the two phases. Therefore, we choose the alphabet $\mathcal{V}_i$ of the feedback V_i (see Figure 1) to be of size 1 if $i \neq \alpha n$ and of size $2^{nR_{\text{FB}}}$ if $i = \alpha n$. Of course, one could use the feedback-link more evenly and make sure that each alphabet $\mathcal{V}_i$ is of size at most $2^{R_{\text{FB}}}$ by using B instances of the two phases, where $B \to \infty$, and by starting the transmission with sufficiently many instances of Phase 1.

receiver $\mathcal{Z}$'s compression index consists of $\alpha n H(Z|Y)$ bits, where the conditional entropy is w.r.t. a PMF $p(u,x,y,z)$ of the form (30) for which the rate-tuple $(\tilde{R}_{\mathcal{Y}}^{(p)}, 0, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)})$ satisfies (29) ahead. Thus, the compression index can be sent over the rate-limited feedback link if

$$\alpha H(Z|Y) \leq R_{\text{FB}},$$

and the encoder can send it at rate $R_{\mathcal{Y}}^{(c)}$ in Phase 2 if

$$\alpha H(Z|Y) \leq (1-\alpha) R_{\mathcal{Y}}^{(c)}.$$

At the end of each phase, the stochastic receiver $\mathcal{Z}$ decodes the message sent to it during that phase. The deterministic receiver $\mathcal{Y}$ waits with decoding until the end of Phase 2, when it first decodes its intended phase-2 message and the compression index. From the compression index it estimates receiver $\mathcal{Z}$'s phase-1 channel outputs. It then decodes its intended phase-1 message based on its phase-1 channel outputs and its estimate of receiver $\mathcal{Z}$'s phase-1 channel outputs. The described code achieves the rate-tuple

$$\alpha(\tilde{R}_{\mathcal{Y}}^{(p)}, 0, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}) + (1-\alpha)(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}). \tag{26}$$

If also the stochastic receiver $\mathcal{Z}$ has P-MSI, then the above code needs to be modified in the following way: in Phase 1 the encoder must code for the enhanced BC with P-MSI at both receivers, and in Phase 2 the compression index constitutes additional P-MSI at the stochastic receiver, i.e., P-MSI beyond what it already had in the beginning. In Phase 1 the encoder thus sends at rates $(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)})$ satisfying (29) ahead for some fixed PMF $p(u,x,y,z)$ of the form (30), and in Phase 2 the encoder sends fresh message-information at rates $(R_{\mathcal{Y}}^{(p)}, \hat{R}_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$, where

$$\hat{R}_{\mathcal{Y}}^{(c)} \leq R_{\mathcal{Y}}^{(c)} - \frac{\alpha}{1-\alpha} H(Z|Y) m \tag{27}$$

and where $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}$. Hence, if also receiver $\mathcal{Z}$ has P-MSI, then the described code achieves the rate-tuple

$$\alpha(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Y}}^{(c)}, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}) + (1-\alpha)(R_{\mathcal{Y}}^{(p)}, \hat{R}_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}). \tag{28}$$

A detailed characterization of the rate-tuples that are achievable by the above feedback coding scheme is given in Proposition 12 ahead. To state our result, we use the capacity region $\mathscr{C}_{\text{P-MSI}}^{(\text{enh})}$ of the enhanced BC with P-MSI. By construction, the enhanced BC is degraded, and its capacity region can be derived from [9, Theorem 3].

Proposition 10 (Enhanced BC with P-MSI)**.** *The capacity region $\mathscr{C}_{P\text{-}MSI}^{(enh)}$ of the enhanced BC with P-MSI is the set of rate-tuples $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying*

$$R_{\mathcal{Z}} \leq I(U;Z) \tag{29a}$$

$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \leq I(X;Y,Z) \tag{29b}$$

$$R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}} \leq I(X;Y,Z|U) + I(U;Z) \tag{29c}$$

for some PMF of the form

$$p(u,x,y,z) = p(u,x)W(y,z|x). \tag{30}$$

Proof. The result is an immediate consequence of [9, Theorem 3], which is a characterization of the capacity region of the BC with P-MSI and degraded message sets. The reason for this is that the enhanced BC is a degraded BC whose stronger receiver is $\mathcal{Y}$, and that, as we argue next, on the degraded BC we can w.l.g. assume that the stronger receiver $\mathcal{Y}$ must also decode the weaker receiver $\mathcal{Z}$'s message $M_{\mathcal{Z}}^{(p)}$. Indeed, since the weaker receiver $\mathcal{Z}$ can decode $(M_{\mathcal{Z}}^{(p)}, M_{\mathcal{Z}}^{(c)})$ from the outputs that it observes and $M_{\mathcal{Y}}^{(c)}$, and since receiver $\mathcal{Y}$ can decode $(M_{\mathcal{Y}}^{(p)}, M_{\mathcal{Y}}^{(c)})$ from the outputs that it observes and $M_{\mathcal{Z}}^{(c)}$, it follows that the stronger receiver $\mathcal{Y}$ can decode $M_{\mathcal{Z}}^{(p)}$ from the outputs that it observes and $(M_{\mathcal{Y}}^{(p)}, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$. □

Note that on the enhanced BC P-MSI at receiver $\mathcal{Y}$ only cannot increase capacity:

Note 11 (Enhanced BC without MSI at $\mathcal{Z}$)**.** *Denote by $\mathscr{C}^{(enh)}$ the capacity region of the enhanced SD-BC without MSI, i.e., the set of rate-pairs $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ satisfying*

$$R_{\mathcal{Y}} \leq I(X;Y,Z|U) \tag{31a}$$

$$R_{\mathcal{Z}} \leq I(U;Z) \tag{31b}$$

for some PMF of the form (30)*. When receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$), then irrespective of $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$ the capacity region of the enhanced BC is the set of rate-tuples $(R_{\mathcal{Y}}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying* (31) *for some PMF of the form* (30)*.*

With the capacity region of the enhanced BC at hand, we are now ready to characterize the rate-tuples that are achievable by the described feedback code:

Proposition 12 (Feedback Code)**.** *Fix a PMF $p(u,x,y,z)$ of the form* (30)*, and let $(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Y}}^{(c)}, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)})$ be a rate-tuple that satisfies* (29) *for the fixed PMF. Also, pick a rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{P\text{-}MSI}$, a nonnegative number*

$$\alpha \leq \min\left\{ \frac{R_{\mathcal{Y}}^{(c)}}{R_{\mathcal{Y}}^{(c)} + H(Z|Y)}, \frac{R_{FB}}{H(Z|Y)} \right\}, \tag{32}$$

where the conditional entropy $H(Z|Y)$ is evaluated for the fixed PMF, and a nonegative rate

$$\hat{R}_{\mathcal{Y}}^{(c)} \leq R_{\mathcal{Y}}^{(c)} - \frac{\alpha}{1-\alpha} H(Z|Y). \tag{33}$$

The capacity region of the SD-BC with P-MSI and rate-limited feedback of rate R_{FB} from the stochastic receiver $\mathcal{Z}$ contains the rate-tuple

$$\alpha(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Y}}^{(c)}, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}) + (1-\alpha)(R_{\mathcal{Y}}^{(p)}, \hat{R}_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}). \tag{34}$$

Proof. See Appendix E. □

Using Proposition 12, we next identify sufficient conditions for feedback to increase the capacity of the SD-BC. For simplicity, we assume that receiver $\mathcal{Z}$ has no MSI. Proposition 13 treats the case with P-MSI at receiver $\mathcal{Y}$ and Proposition 14 the case without MSI.

Proposition 13 (Sufficient Conditions with P-MSI at $\mathcal{Y}$)**.** *Consider an SD-BC where only the deterministic receiver $\mathcal{Y}$ has MSI ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$). If there exists a rate-triple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying*

$$(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)}) \in \big(\partial\mathscr{C} \cap (\mathscr{C}_{enh} \setminus \partial\mathscr{C}_{enh})\big), \tag{35}$$

and if for some PMF $p(u,x,y,z)$ of the form (12) *Conditions* (11) *and*

$$R_{\mathcal{Y}} < H(Y) \quad \text{and} \quad 0 < I(U;Y) \tag{36}$$

hold, then, irrespective of $R_{FB} > 0$, $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ is in the interior of the feedback capacity region, and feedback thus increases capacity.

Proof. See Appendix F. □

Proposition 14 (Sufficient Conditions without MSI)**.** *Consider an SD-BC without MSI ($R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$). If there exists a rate-pair $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ satisfying*

$$(R_{\mathcal{Y}}, R_{\mathcal{Z}}) \in \big(\partial\mathscr{C} \cap (\mathscr{C}_{enh} \setminus \partial\mathscr{C}_{enh})\big), \tag{37}$$

and if for some PMF $p(u,v,x,y,z)$ of the form (2) *Conditions* (4) *and*

$$R_{\mathcal{Y}} < H(Y) \quad \text{and} \quad 0 < I(V;Y) - I(V;Z) \tag{38}$$

hold, then, irrespective of $R_{FB} > 0$, $(R_{\mathcal{Y}}, R_{\mathcal{Z}})$ is in the interior of the feedback capacity region, and feedback thus increases capacity.

Proof. See Appendix F. □

Proposition 14 can also be derived from the coding scheme of Wu and Wigger [5]:

Remark 1 (Feedback Code of [5])**.** *Let $Y_1 = Z$ and $Y_2 = Y$, and call receiver $\mathcal{Z}$ receiver 1 and receiver $\mathcal{Y}$ receiver 2. If for some PMF $p(v, u, x, y, z)$ of the form* (2) *for which $0 < I(V;Y) - I(V;Z)$ we set $U_0 = V$, $U_1 = U$, and $U_2 = Y$ in Marton's inner bound [14, Prop. 8.1], then Proposition 14 follows from [5, Corollary 3].*

Proposition 14 is used in the analysis of Example 4 ahead (see Appendix G), which proves that feedback can increase the capacity of the SD-BC without MSI. In this analysis, one chooses $U = V$, which turns Marton coding into the simpler superposition coding without satellite for Receiver $\mathcal{Z}$. There are other SD-BCs where one has to apply Proposition 14 with full Marton coding, i.e., with $V \neq U$, to show that feedback increases capacity (see Remark 3 in Appendix G).

It is surprising that Proposition 14 can be used to show that feedback increases the capacity of the SD-BC without MSI: The boundary points of the no-feedback capacity region of the SD-BC without MSI are known to be achievable using only satellites but no cloud-center, i.e., $V = \emptyset$. In contrast, Proposition 14 relies on the assumption that one can achieve boundary points using full Marton coding with a cloud-center, i.e., $V \neq \emptyset$ (see (38)).

4.2 Main Results

Using Proposition 12 (or, more precisely, the sufficient conditions of Propositions 13 and 14 that we derived from Proposition 12), we show that feedback can increase the capacity of the SD-BC with and without P-MSI. Key to the proof are the following two observations:

1. MSI to the stochastic receiver can increase the capacity of the SD-BC.
2. The capacity region of the enhanced BC with P-MSI is typically larger than that of the SD-BC with P-MSI.

Whether feedback can increase the capacity of the SD-BC with P-MSI depends on the nature of the P-MSI. If the stochastic receiver $\mathcal{Z}$ has no MSI, then feedback can increase capacity:

Theorem 15 (Feedback Helps)**.** *When the stochastic receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$), then rate-limited feedback of any positive rate can increase the capacity region*

of the SD-BC irrespective of whether or not the deterministic receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$). It can in particular increase the sum-rate capacity when receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$).

The theorem follows from the following example, which we analyze in Appendix G:

Example 4 (Feedback Helps)**.** *On the SD-BC with binary inputs X_1 and X_2, binary state $S \sim \mathrm{Ber}(p)$, $p \in (0,1)$, and outputs $Y = X_1 + X_2$ and*

$$Z = \begin{cases} X_2 & S = 0, \\ ? & S = 1, \end{cases}$$

rate-limited feedback of any positive rate from the stochastic receiver $\mathcal{Z}$ increases the sum-rate capacity when receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$) and receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$). It increases the capacity region when $p > 1/2$ and no receiver has MSI ($R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$ and $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$).

The setting where the stochastic receiver has F-MSI is different:

Theorem 16 (Feedback with F-MSI at $\mathcal{Z}$)**.** *When the stochastic receiver $\mathcal{Z}$ has F-MSI ($R_{\mathcal{Y}}^{(p)} = 0$), even perfect feedback cannot increase the capacity region of the SD-BC. This holds irrespective of whether or not receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$).*

Proof. See Appendix H. □

Of course, whether feedback can increase the capacity of the SD-BC also depends on the channel. The following is an example of an SD-BC that does not consist of two noninterfering single-user channels, and that is neither deterministic nor physically-degraded, but still feedback does not increase its capacity without MSI at receiver $\mathcal{Z}$, irrespective of whether or not receiver $\mathcal{Y}$ has P-MSI. The example is analyzed in Appendix I.

Example 5 (Feedback with F-MSI at $\mathcal{Z}$)**.** *Consider the SD-BC with binary state $S \sim \mathrm{Ber}(p)$, $p \in (0,1)$, and outputs $Y = f(X)$ and*

$$Z = \begin{cases} X & \text{if } S = 0, \\ ? & \text{if } S = 1, \end{cases} \tag{39}$$

where $\mathcal{Y} \subsetneq \mathcal{X}$ and $? \notin \mathcal{X}$. Denote $\bar{p} = 1 - p$, and for every $y \in \mathcal{Y}$ define $\mathcal{X}_y \triangleq \{x \in \mathcal{X} \colon f(x) = y\}$. When the stochastic receiver $\mathcal{Z}$ has no MSI ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$), then

irrespective of $R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$ *the capacity region of this SD-BC is the set of rate-tuples* $(R_{\mathcal{Y}}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ *satisfying*

$$R_{\mathcal{Y}} \leq H(Y) \tag{40a}$$

$$R_{\mathcal{Z}} \leq \bar{p} I(U;Y) + \bar{p} \sum_{y \in \mathcal{Y}} p(y) \log|\mathcal{X}_y| \tag{40b}$$

$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq H(Y) - pI(U;Y) + \bar{p} \sum_{y \in \mathcal{Y}} p(y) \log|\mathcal{X}_y| \tag{40c}$$

for some PMF $p(u, x, y, z)$ *of the form* (12) *satisfying that* X *is a function of* (Y, U)*. This also holds in the presence of feedback, and thus feedback cannot increase the capacity region.*

Similar results as in Theorems 15 and 16 hold if the SD-BC is state-dependent and the encoder is furnished with strictly-causal state information (SI) instead of feedback:

Theorem 17 (Strictly-Causal SI)**.** *Suppose that the SD-BC is governed by an i.i.d. state-sequence of finite support* $\mathcal{S}$*, i.e.,*

$$W(y, z|x, s) = \mathbb{1}_{\{y = f(x)\}} W(z|x, s), \; s \in \mathcal{S}.$$

When the stochastic receiver $\mathcal{Z}$ *has no MSI* ($R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$)*, strictly-causal SI can increase the capacity of the SD-BC irrespective of whether or not the deterministic receiver* $\mathcal{Y}$ *has P-MSI* ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$)*, and it can in particular increase the sum-rate capacity when receiver* $\mathcal{Y}$ *has P-MSI* ($R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$)*. When the stochastic receiver* $\mathcal{Z}$ *has F-MSI* ($R_{\mathcal{Y}}^{(p)} = 0$)*, strictly-causal SI cannot increase capacity irrespective of whether or not receiver* $\mathcal{Y}$ *has P-MSI* ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$)*.*

Proof. Note that the functional representation lemma (Lemma 1) allows us to view an SD-BC $W(y, z|x)$ as a state-dependent SD-BC $W(y, z|x, s)$ whose stochastic output Z can be computed from its input and state. If the state is revealed strictly-causally to the encoder, then the encoder can compute the past channel outputs at the stochastic receiver $\mathcal{Z}$ from the past states and the past channel inputs. Hence, on the SD-BC feedback is no better than strictly-causal SI, and thus we obtain the first claim as a corollary to Theorem 15.

To see that when the stochastic receiver $\mathcal{Z}$ has F-MSI ($R_{\mathcal{Y}}^{(p)} = 0$) strictly-causal SI cannot increase capacity irrespective of whether or not receiver $\mathcal{Y}$ has P-MSI ($R_{\mathcal{Z}}^{(p)} \in [0, R_{\mathcal{Z}}]$), one can follow the line of arguments in the proof of Theorem 16. □

4.3 Intuition on the Results with Feedback

Based on our coding scheme in Section 4.1, we can offer the following intuition: feedback on the SD-BC (with or without P-MSI) can increase capacity because it allows the encoder to build up P-MSI at the stochastic receiver and thus allows the stochastic receiver to better mitigate experienced interference (see Note 5 and the discussion in Section 3.3). The setting where the stochastic receiver has F-MSI and feedback cannot increase capacity is different: F-MSI suffices for the stochastic receiver to fully mitigate experienced interference, and thus feedback cannot provide further benefits.

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A A Proof of Theorem 2

A.1 A Direct Part

Codebook Generation: Fix numbers $\tilde{\epsilon} > \epsilon > 0$, a PMF $p(v,u,x) = p(v,u)p(x|u)$, and $\tilde{R}^{(c)}$, $\tilde{R}_{\mathcal{Y}}$, $\tilde{R}_{\mathcal{Z}} > 0$ such that $\tilde{R}^{(c)} \leq \min\{\tilde{R}_{\mathcal{Y}}, \tilde{R}_{\mathcal{Z}}\}$. For each $(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)}) \in \mathcal{M} \times \mathcal{M}_{\mathcal{Y}}^{(c)} \times \mathcal{M}_{\mathcal{Z}}^{(c)}$ draw $2^{n\tilde{R}^{(c)}}$ n-tuples $\mathbf{v}$ from the product distribution $\prod_{i=1}^{n} p(v_i)$, label them by $k \in [1 : 2^{n\tilde{R}^{(c)}}]$, where $\mathbf{v}(k)$ denotes the n-tuple labelled by k, and place them in a bin $\mathscr{B}(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)})$. For each n-tuple $\mathbf{v}$ draw $2^{n(\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)})}$ n-tuples $\mathbf{y} \in \mathcal{Y}^n$ independently of each other and each from the product distribution $\prod_{i=1}^{n} p(y_i|v_i)$, and label them by $\ell \in [1 : 2^{n(\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)})}]$, where $\mathbf{y}(k,\ell)$ denotes the ℓ-th n-tuple corresponding to $\mathbf{v}(k)$. Randomly allocate the $2^{n\tilde{R}_{\mathcal{Y}}}$ n-tuples to $2^{nR_{\mathcal{Y}}^{(p)}}$ bins $\mathscr{B}_{\mathcal{Y}}(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)}, m_{\mathcal{Y}}^{(p)})$, where $m_{\mathcal{Y}}^{(p)} \in \mathcal{M}_{\mathcal{Y}}^{(p)}$. Similarly, for each n-tuple $\mathbf{v}$ draw $2^{n(\tilde{R}_{\mathcal{Z}} - \tilde{R}^{(c)})}$ n-tuples $\mathbf{u} \in \mathcal{U}^n$ independently of each other and each from the product distribution $\prod_{i=1}^{n} p(u_i|v_i)$, and label them by $j \in [1 : 2^{n(\tilde{R}_{\mathcal{Z}} - \tilde{R}^{(c)})}]$, where $\mathbf{u}(k,j)$ denotes the j-th n-tuple $\mathbf{u}$ corresponding to $\mathbf{v}(k)$. Randomly allocate the $2^{n\tilde{R}_{\mathcal{Z}}}$ n-tuples to $2^{nR_{\mathcal{Z}}^{(p)}}$ bins $\mathscr{B}_{\mathcal{Z}}(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)}, m_{\mathcal{Z}}^{(p)})$, where $m_{\mathcal{Z}}^{(p)} \in \mathcal{M}_{\mathcal{Z}}^{(p)}$.

Encoding: If there are n-tuples $\mathbf{v} \in \mathscr{B}(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)})$, $\mathbf{y} \in \mathscr{B}_{\mathcal{Y}}(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, M_{\mathcal{Y}}^{(p)})$, and $\mathbf{u} \in \mathscr{B}_{\mathcal{Z}}(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, M_{\mathcal{Z}}^{(p)})$ such that $(\mathbf{v}, \mathbf{y}, \mathbf{u}) \in \mathcal{T}_{\epsilon}^{(n)}(V, Y, U)$, then the encoder draws the length-n channel input sequence X^n from the product distribution $\prod_{i=1}^{n} p(x_i|y_i, u_i)$. Otherwise the encoding is unsuccessful.

Decoding: Upon observing $\left(Y^n, M_{\mathcal{Z}}^{(c)}\right)$, receiver $\mathcal{Y}$ decodes $(m, m_{\mathcal{Y}})$ if it is the unique element of $\mathcal{M} \times \mathcal{M}_{\mathcal{Y}}$ such that $Y^n \in \mathscr{B}_{\mathcal{Y}}(m, m_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, m_{\mathcal{Y}}^{(p)}) \cap \mathcal{T}_{\tilde{\epsilon}}^{(n)}(Y)$. Otherwise it declares an error. Upon observing $\left(Z^n, M_{\mathcal{Y}}^{(c)}\right)$, receiver $\mathcal{Z}$ decodes $(m, m_{\mathcal{Z}})$ if it is the

unique element of $\mathcal{M}_{\mathcal{Z}}$ such that Bin $\mathscr{B}_{\mathcal{Z}}(m, M_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)}, m_{\mathcal{Z}}^{(p)})$ contains an n-tuple $\mathbf{u}$ satisfying $(\mathbf{u}, Z^n) \in \mathcal{T}_{\tilde{\epsilon}}^{(n)}(U, Z)$. Otherwise it declares an error.

Analysis of the Error Probability: We first analyze the probability that the encoding is unsuccessful. By symmetry, it does not depend on the messages' realizations, and we thus assume w.l.g. that $M = M_{\mathcal{Y}}^{(p)} = M_{\mathcal{Y}}^{(c)} = M_{\mathcal{Z}}^{(p)} = M_{\mathcal{Z}}^{(c)} = 1$. Denote $\mathscr{B} \triangleq \mathscr{B}(1,1,1)$, $\mathscr{B}_{\mathcal{Y}} \triangleq \mathscr{B}_{\mathcal{Y}}(1,1,1,1)$, and $\mathscr{B}_{\mathcal{Z}} \triangleq \mathscr{B}_{\mathcal{Z}}(1,1,1,1)$. Note that the encoder can choose one out of

$$N = \sum_{k,\ell,j} \mathbb{1}_{(\mathbf{v}(k),\mathbf{y}(k,\ell),\mathbf{u}(k,j)) \in \mathcal{T}_{\epsilon}^{(n)}} \mathbb{1}_{\mathbf{y}(k,\ell) \in \mathscr{B}_{\mathcal{Y}}} \mathbb{1}_{\mathbf{u}(k,j) \in \mathscr{B}_{\mathcal{Z}}}$$

triples $(\mathbf{v}, \mathbf{y}, \mathbf{u})$, where $k \in [1 : 2^{n\tilde{R}^{(c)}}]$, $\ell \in [1 : 2^{n(\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)})}]$, and $j \in [1 : 2^{n(\tilde{R}_{\mathcal{Z}} - \tilde{R}^{(c)})}]$. The encoding is unsuccessful if $N = 0$. As in the proof of the mutual covering lemma [14, Lemma 8.1], we obtain from Chebyshev's inequality that

$$\mathbb{P}[N = 0] \leq \mathbb{P}\big[\big(N - \mathbb{E}[N]\big)^2 \geq \mathbb{E}[N]^2\big] \leq \frac{\mathrm{Var}(N)}{\mathbb{E}[N]^2}. \tag{41}$$

To conclude that on average over the realization of the code the encoding is with high probability successful, it thus suffices to show that

$$\frac{\mathrm{Var}(N)}{\mathbb{E}[N]^2}$$

converges to 0 as n tends to infinity. Define

$$E^{k,\ell,j} \triangleq \mathbb{1}_{(\mathbf{v}(k),\mathbf{y}(k,\ell),\mathbf{u}(k,j)) \in \mathcal{T}_{\epsilon}^{(n)}} \mathbb{1}_{\mathbf{y}(k,\ell) \in \mathscr{B}_{\mathcal{Y}}} \mathbb{1}_{\mathbf{u}(k,j) \in \mathscr{B}_{\mathcal{Z}}} \tag{42a}$$

$$T^{k,\ell,j} \triangleq \mathbb{1}_{(\mathbf{v}(k),\mathbf{y}(k,\ell),\mathbf{u}(k,j)) \in \mathcal{T}_{\epsilon}^{(n)}}, \tag{42b}$$

and note that

$$\begin{aligned}
\mathbb{E}[N^2] &= \sum_{k,\ell,j} \sum_{k',\ell',j'} \mathbb{E}\big[E^{k,\ell,j} E^{k',\ell',j'}\big] && (43)\\
&= \sum_{k,\ell,j} \sum_{k' \neq k,\ell',j'} \mathbb{E}\big[E^{k,\ell,j} E^{k',\ell',j'}\big] \\
&\quad + \sum_{k,\ell,j} \sum_{\ell' \neq \ell, j' \neq j} \mathbb{E}\big[E^{k,\ell,j} E^{k,\ell',j'}\big] \\
&\quad + \sum_{k,\ell,j} \sum_{j' \neq j} \mathbb{E}\big[E^{k,\ell,j} E^{k,\ell,j'}\big] \\
&\quad + \sum_{k,\ell,j} \sum_{\ell' \neq \ell} \mathbb{E}\big[E^{k,\ell,j} E^{k,\ell',j}\big] \\
&\quad + \sum_{k,\ell,j} \mathbb{E}\big[E^{k,\ell,j}\big], && (44)
\end{aligned}$$

where we used that $E^{k,\ell,j}E^{k,\ell,j} = E^{k,\ell,j}$. If $k \neq k'$, then $E^{k,\ell,j}$ and $E^{k',\ell',j'}$ are independent and

$$\begin{aligned}
&\sum_{k,\ell,j}\sum_{k'\neq k,\ell',j'} \mathbb{E}\big[E^{k,\ell,j}E^{k',\ell',j'}\big] \\
&\quad = \sum_{k,\ell,j}\sum_{k'\neq k,\ell',j'} \mathbb{E}\big[E^{k,\ell,j}\big]\,\mathbb{E}\big[E^{k',\ell',j'}\big] && (45)\\
&\quad \leq \left(\sum_{k,\ell,j}\mathbb{E}\big[E^{k,\ell,j}\big]\right)^2 = \mathbb{E}[N]^2. && (46)
\end{aligned}$$

If $k = k'$, $\ell \neq \ell'$, and $j \neq j'$, then

$$\begin{aligned}
&\sum_{k,\ell,j}\sum_{\ell'\neq \ell,j'\neq j} \mathbb{E}\big[E^{k,\ell,j}E^{k,\ell',j'}\big] \\
&\quad \overset{(a)}{\leq} 2^{-2n(R_{\mathcal{Y}}^{(p)}+R_{\mathcal{Z}}^{(p)})}\sum_{k,\ell,j}\sum_{\ell'\neq \ell,j'\neq j}\mathbb{E}\big[T^{k,\ell,j}T^{k,\ell',j'}\big] && (47)\\
&\quad \overset{(b)}{\leq} 2^{-2n(R_{\mathcal{Y}}^{(p)}+R_{\mathcal{Z}}^{(p)})}\sum_{k,\ell,j}\sum_{\ell'\neq \ell,j'\neq j} 2^{-n(2I(Y;U|V)-\delta(\epsilon))} && (48)\\
&\quad = 2^{n(2\tilde{R}_{\mathcal{Y}}-2R_{\mathcal{Y}}^{(p)}+2\tilde{R}_{\mathcal{Z}}-2R_{\mathcal{Z}}^{(p)}-3\tilde{R}^{(c)}-2I(Y;U|V)+\delta(\epsilon))}, && (49)
\end{aligned}$$

where (a) is true because $\mathbb{1}_{\mathbf{y}(k,\ell)\in\mathscr{B}_{\mathcal{Y}}}$, $\mathbb{1}_{\mathbf{u}(k,j)\in\mathscr{B}_{\mathcal{Z}}}$, $\mathbb{1}_{\mathbf{y}(k,\ell')\in\mathscr{B}_{\mathcal{Y}}}$, and $\mathbb{1}_{\mathbf{u}(k,j')\in\mathscr{B}_{\mathcal{Z}}}$ are independent of each other and of $T^{k,\ell,j}T^{k,\ell',j'}$, and because

$$\begin{aligned}
\mathbb{E}\big[\mathbb{1}_{\mathbf{y}(k,\ell)\in\mathscr{B}_{\mathcal{Y}}}\big] &= 2^{-nR_{\mathcal{Y}}^{(p)}}\\
\mathbb{E}\big[\mathbb{1}_{\mathbf{u}(k,j)\in\mathscr{B}_{\mathcal{Z}}}\big] &= 2^{-nR_{\mathcal{Z}}^{(p)}},
\end{aligned}$$

and where (b) holds because of the properties of typical sequences. (Recall that $\delta(\cdot)$ denotes some function of ϵ that converges to 0 as ϵ approaches 0.) If $k = k'$, $\ell = \ell'$, and $j \neq j'$, then

$$\begin{aligned}
&\sum_{k,\ell,j}\sum_{j'\neq j} \mathbb{E}\big[E^{k,\ell,j}E^{k,\ell,j'}\big] \\
&\quad \overset{(a)}{\leq} 2^{-n(R_{\mathcal{Y}}^{(p)}+2R_{\mathcal{Z}}^{(p)})}\sum_{k,\ell,j}\sum_{j'\neq j}\mathbb{E}\big[T^{k,\ell,j}T^{k,\ell,j'}\big] && (50)\\
&\quad \overset{(b)}{\leq} 2^{-n(R_{\mathcal{Y}}^{(p)}+2R_{\mathcal{Z}}^{(p)})}\sum_{k,\ell,j}\sum_{j'\neq j} 2^{-n(2I(Y;U|V)-\delta(\epsilon))} && (51)\\
&\quad = 2^{n(\tilde{R}_{\mathcal{Y}}-R_{\mathcal{Y}}^{(p)}+2\tilde{R}_{\mathcal{Z}}-2R_{\mathcal{Z}}^{(p)}-2\tilde{R}^{(c)}-2I(Y;U|V)+\delta(\epsilon))}, && (52)
\end{aligned}$$

where (a) holds since $\mathbb{1}_{\mathbf{y}(k,\ell)\in\mathscr{B}_{\mathcal{Y}}}\mathbb{1}_{\mathbf{y}(k,\ell)\in\mathscr{B}_{\mathcal{Y}}} = \mathbb{1}_{\mathbf{y}(k,\ell)\in\mathscr{B}_{\mathcal{Y}}}$ and since $\mathbb{1}_{\mathbf{y}(k,\ell)\in\mathscr{B}_{\mathcal{Y}}}$, $\mathbb{1}_{\mathbf{u}(k,j)\in\mathscr{B}_{\mathcal{Z}}}$, and $\mathbb{1}_{\mathbf{u}(k,j')\in\mathscr{B}_{\mathcal{Z}}}$ are independent of each other and of $T^{k,\ell,j}T^{k,\ell,j'}$, and where (b) follows from the properties of typical sequences. Similarly, if $k = k'$, $j = j'$, and $\ell \neq \ell'$, then

$$\sum_{k,\ell,j}\sum_{\ell'\neq\ell}\mathbb{E}\left[E^{k,\ell,j}E^{k,\ell',j}\right] = 2^{n(2\tilde{R}_{\mathcal{Y}}-2R_{\mathcal{Y}}^{(p)}+\tilde{R}_{\mathcal{Z}}-R_{\mathcal{Z}}^{(p)}-2\tilde{R}^{(c)}-2I(Y;U|V)+\delta(\epsilon))}. \tag{53}$$

If $k = k'$, $j = j'$, and $\ell = \ell'$, then

$$\sum_{k,\ell,j}\mathbb{E}\left[E^{k,\ell,j}\right] = \mathbb{E}[N]. \tag{54}$$

Using that $\mathrm{Var}(N) = \mathbb{E}[N^2] - \mathbb{E}[N]^2$, we obtain from (41), (44), (46), (49), (52), (53), and (54) that $\mathbb{P}[N = 0]$ converges to 0 as n tends to infinity if asymptotically the RHS of each equation (49), (52), (53), and (54) is negligibly small compared to $\mathbb{E}[N]^2$. Note that

$$\mathbb{E}[N] = \sum_{k,\ell,j}\mathbb{E}\left[E^{k,\ell,j}\right] \tag{55}$$

$$\overset{(a)}{=} 2^{-n(R_{\mathcal{Y}}^{(p)}+R_{\mathcal{Z}}^{(p)})}\sum_{k,\ell,j}\mathbb{E}\left[T^{k,\ell,j}\right] \tag{56}$$

$$\overset{(b)}{\geq} 2^{-n(R_{\mathcal{Y}}^{(p)}+R_{\mathcal{Z}}^{(p)})}\sum_{k,\ell,j}2^{-n(I(Y;U|V)+\delta(\epsilon))} \tag{57}$$

$$= 2^{n(\tilde{R}_{\mathcal{Y}}-R_{\mathcal{Y}}^{(p)}+\tilde{R}_{\mathcal{Z}}-R_{\mathcal{Z}}^{(p)}-\tilde{R}^{(c)}-I(Y;U|V)-\delta(\epsilon))}, \tag{58}$$

where (a) is true because $\mathbb{1}_{\mathbf{y}(k,\ell)\in\mathscr{B}_{\mathcal{Y}}}$ and $\mathbb{1}_{\mathbf{u}(k,j)\in\mathscr{B}_{\mathcal{Z}}}$ are independent of each other and of $T^{k,\ell,j}$, and where (b) holds for all sufficiently large n by the properties of typical sequences. Hence, on average over the realization of the code the probability that the encoding is unsuccessful converges to 0 as n tends to infinity if

$$\tilde{R}^{(c)} > 3\delta(\epsilon) \tag{59a}$$

$$\tilde{R}_{\mathcal{Y}} - R_{\mathcal{Y}}^{(p)} > 3\delta(\epsilon) \tag{59b}$$

$$\tilde{R}_{\mathcal{Z}} - R_{\mathcal{Z}}^{(p)} > 3\delta(\epsilon) \tag{59c}$$

$$\tilde{R}_{\mathcal{Y}} - R_{\mathcal{Y}}^{(p)} + \tilde{R}_{\mathcal{Z}} - R_{\mathcal{Z}}^{(p)} - \tilde{R}^{(c)} > I(Y;U|V) + \delta(\epsilon). \tag{59d}$$

Assuming that the encoding was successful, we now analyze the probability that the decoding is unsuccessful. Since receiver $\mathcal{Y}$ observes the n-tuple $\mathbf{y}$ that the encoder selected and because $\mathcal{T}_{\epsilon}^{(n)}(Y) \subseteq \mathcal{T}_{\tilde{\epsilon}}^{(n)}(Y)$, it is clear that $Y^n \in \mathscr{B}_{\mathcal{Y}}(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, M_{\mathcal{Y}}^{(p)}) \cap$

$\mathcal{T}_{\tilde{\epsilon}}^{(n)}(Y)$. Hence, the pair $(M, M_{\mathcal{Y}})$ satisfies the decoding requirements. Moreover, for all sufficiently large n it holds with high probability that $(U^n, Z^n) \in \mathcal{T}_{\tilde{\epsilon}}^{(n)}(P_{U,Z})$ and thus that the pair $(M, M_{\mathcal{Z}})$ satisfies the decoding requirements. Therefore, we need only worry about the event that also other message-pairs $(m, m_{\mathcal{Y}}) \in \mathcal{M} \times \mathcal{M}_{\mathcal{Y}} \setminus \{(M, M_{\mathcal{Y}})\}$ or $(m, m_{\mathcal{Z}}) \in \mathcal{M} \times \mathcal{M}_{\mathcal{Z}} \setminus \{(M, M_{\mathcal{Z}})\}$ satisfy the decoding requirements. For receiver $\mathcal{Y}$ this can happen if an n-tuple $\mathbf{y}$ that corresponds to the same length-n sequence $\mathbf{v}$ as Y^n satisfies the decoding requirements, or if an n-tuple $\mathbf{y}$ that corresponds to a different length-n sequence $\mathbf{v}$ than Y^n satisfies the decoding requirements, and similarly for receiver $\mathcal{Z}$. Hence, by the properties of typical sequences the probability that the decoding is unsuccessful converges to 0 as n tends to infinity if

$$\tilde{R}_{\mathcal{Y}} - \tilde{R}^{(c)} < H(Y|V) - \delta(\tilde{\epsilon}) \tag{60a}$$

$$\tilde{R}_{\mathcal{Y}} + R + R_{\mathcal{Y}}^{(c)} < H(Y) - \delta(\tilde{\epsilon}) \tag{60b}$$

$$\tilde{R}_{\mathcal{Z}} - \tilde{R}^{(c)} < I(U;Z|V) - \delta(\tilde{\epsilon}) \tag{60c}$$

$$\tilde{R}_{\mathcal{Z}} + R + R_{\mathcal{Z}}^{(c)} < I(U;Z) - \delta(\tilde{\epsilon}). \tag{60d}$$

Now let ϵ and $\tilde{\epsilon}$ tend to 0. Then, also $\delta(\epsilon)$ and $\delta(\tilde{\epsilon})$ tend to 0, and we conclude that a rate-tuple $(R, R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ is achievable if there exist $\tilde{R}^{(c)}$, $\tilde{R}_{\mathcal{Y}}$, and $\tilde{R}_{\mathcal{Z}}$ such that

$$-\tilde{R}^{(c)} < 0 \tag{61a}$$

$$R_{\mathcal{Y}}^{(p)} - \tilde{R}_{\mathcal{Y}} < 0 \tag{61b}$$

$$R_{\mathcal{Z}}^{(p)} - \tilde{R}_{\mathcal{Z}} < 0 \tag{61c}$$

$$R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}}^{(p)} + \tilde{R}^{(c)} - \tilde{R}_{\mathcal{Y}} - \tilde{R}_{\mathcal{Z}} < -I(Y;U|V) \tag{61d}$$

$$\tilde{R}^{(c)} - \tilde{R}_{\mathcal{Y}} < 0 \tag{61e}$$

$$-\tilde{R}^{(c)} + \tilde{R}_{\mathcal{Y}} < H(Y|V) \tag{61f}$$

$$R + R_{\mathcal{Y}}^{(c)} + \tilde{R}_{\mathcal{Y}} < H(Y) \tag{61g}$$

$$\tilde{R}^{(c)} - \tilde{R}_{\mathcal{Z}} < 0 \tag{61h}$$

$$-\tilde{R}^{(c)} + \tilde{R}_{\mathcal{Z}} < I(U;Z|V) \tag{61i}$$

$$R + R_{\mathcal{Z}}^{(c)} + \tilde{R}_{\mathcal{Z}} < I(U;Z). \tag{61j}$$

A Fourier-Motzkin elimination reveals that the above inequalities hold if the rate-tuple lies in the interior of $\mathscr{C}_{\text{P-MSI}}$.

At first sight, it may surprise that we also bin the cloud-center. As the following remark shows, this is, however, just an equivalent alternative to rate-splitting. The

benefit is that binning the cloud-center requires only one auxiliary rate, $\tilde{R}^{(c)}$, whereas rate-splitting requires two auxiulary rates, one for each rate $R_{\mathcal{Y}}^{(p)}$ and $R_{\mathcal{Z}}^{(p)}$.

Remark 2. *The effect of binning the cloud-center is that of rate-splitting: generating $2^{n\tilde{R}^{(c)}}$ cloud-centers $\mathbf{v}$ for each triple $(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)})$ is tantamount to dividing the messages $M_{\mathcal{Y}}^{(p)}$ and $M_{\mathcal{Z}}^{(p)}$ into two parts, i.e., $M_{\mathcal{Y}}^{(p)} = (M_{\mathcal{Y},s}^{(p)}, M_{\mathcal{Y},c}^{(p)})$ and $M_{\mathcal{Z}}^{(p)} = (M_{\mathcal{Z},s}^{(p)}, M_{\mathcal{Z},c}^{(p)})$, where we generate for each tuple $(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)}, m_{\mathcal{Y},c}^{(p)}, m_{\mathcal{Z},c}^{(p)})$ a cloud-center $\mathbf{v}$, and where we allocate the associated satellites $\mathbf{y}$ and $\mathbf{u}$ to only $2^{nR_{\mathcal{Y},s}^{(p)}}$ and $2^{nR_{\mathcal{Z},s}^{(p)}}$ rather than $2^{nR_{\mathcal{Y}}^{(p)}}$ and $2^{nR_{\mathcal{Z}}^{(p)}}$ bins, respectively.*

To see that binning the cloud-center is tantamount to rate-splitting, note that with rate-splitting we generate $2^{n(R_{\mathcal{Y},c}^{(p)}+R_{\mathcal{Z},c}^{(p)})}$ cloud-centers $\mathbf{v}$ per triple $(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)})$, namely one for each pair $(m_{\mathcal{Y},c}^{(p)}, m_{\mathcal{Z},c}^{(p)})$. Therefore, we can identify $R_{\mathcal{Y},c}^{(p)}+R_{\mathcal{Z},c}^{(p)}$ in the rate-splitting code with $\tilde{R}^{(c)}$ in the code where we also bin the cloud-center. Moreover, associating every cloud-center $\mathbf{v}$ corresponding to some triple $(m, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(c)})$ with a different pair $(m_{\mathcal{Y},c}^{(p)}, m_{\mathcal{Z},c}^{(p)})$ and allocating the satellites $\mathbf{y}$ and $\mathbf{u}$ to $2^{nR_{\mathcal{Y},s}^{(p)}}$ and $2^{nR_{\mathcal{Z},s}^{(p)}}$ bins, respectively, is tantamount to not associating the cloud-centers with anything but allocating the satellites $\mathbf{y}$ and $\mathbf{u}$ to $2^{nR_{\mathcal{Y}}^{(p)}}$ and $2^{nR_{\mathcal{Z}}^{(p)}}$ bins, respectively. Hence, the only difference between binning the cloud-center and rate-splitting is that for the latter $\tilde{R}^{(c)} = R_{\mathcal{Y},c}^{(p)} + R_{\mathcal{Z},c}^{(p)}$ must satisfy the upper bound $\tilde{R}^{(c)} \leq R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}}^{(p)}$. However, this upper bound is not restrictive: If $\tilde{R}^{(c)} > R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}}^{(p)}$, then we have more cloud-centers $\mathbf{v}$ than message-tuples $(m, m_{\mathcal{Y}}^{(p)}, m_{\mathcal{Y}}^{(c)}, m_{\mathcal{Z}}^{(p)}, m_{\mathcal{Z}}^{(c)})$. Clearly, this cannot be better than having a different cloud-center for each message-tuple.

A.2 A Converse

Let $Q \sim \mathrm{Unif}[1:n]$ be independent of $(M, M_{\mathcal{Y}}, M_{\mathcal{Z}})$, and define

$$V_Q \triangleq \big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Z^{Q-1}\big),\ V \triangleq (V_Q, Q), \tag{62a}$$

$$U_Q \triangleq \big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}\big),\ U \triangleq (U_Q, Q), \tag{62b}$$

$$X \triangleq X_Q,\ Y \triangleq Y_Q,\ Z \triangleq Z_Q. \tag{62c}$$

Note that V, U, X, and (Y, Z) form a Markov chain in that order, i.e., their PMF is of the form (2).

The rate of the message-pair $(M, M_{\mathcal{Y}})$ satisfies

$$R + R_{\mathcal{Y}} - \epsilon_n \overset{(a)}{\leq} \frac{1}{n} I\big(M, M_{\mathcal{Y}}; Y^n, M_{\mathcal{Z}}^{(c)}\big) \tag{63}$$

$$\overset{(b)}{=} I\big(M, M_{\mathcal{Y}}; Y_Q \big| M_{\mathcal{Z}}^{(c)}, Y^{Q-1}, Q\big) \tag{64}$$

$$\overset{(c)}{\leq} H(Y), \tag{65}$$

where (a) follows from Fano's inequality, (b) is due to the chain-rule and the independence of $(M, M_{\mathcal{Y}})$ and $M_{\mathcal{Z}}^{(c)}$, and (c) holds since conditional entropy is nonnegative and conditioning cannot increase entropy. The rate of the message-pair $(M, M_{\mathcal{Z}})$ satisfies

$$R + R_{\mathcal{Z}} - \epsilon_n \overset{(a)}{\leq} \frac{1}{n} I\big(M, M_{\mathcal{Z}}; Z^n, M_{\mathcal{Y}}^{(c)}\big) \tag{66}$$

$$\overset{(b)}{=} I\big(M, M_{\mathcal{Z}}; Z_Q \big| M_{\mathcal{Y}}^{(c)}, Z^{Q-1}, Q\big) \tag{67}$$

$$\overset{(c)}{\leq} I(U; Z), \tag{68}$$

where (a) follows from Fano's inequality, (b) holds because of the chain-rule and since $(M, M_{\mathcal{Z}})$ and $M_{\mathcal{Y}}^{(c)}$ are independent, and (c) is true because conditioning cannot increase entropy.

We prove the sum-rate constraints using Csiszár's sum-identity:

$$\begin{aligned} 0 &= \frac{1}{n} \sum_{i=1}^{n} \big[I(A_{i+1}^n; B_i | B^{i-1}, T) - I(B^{i-1}; A_i | A_{i+1}^n, T) \big] \\ &= I(A_{Q+1}^n; B_Q | B^{Q-1}, T, Q) - I(B^{Q-1}; A_Q | A_{Q+1}^n, T, Q), \end{aligned}$$

where the tuple (A^n, B^n, T) is independent of Q. Note that

$$\begin{aligned} & R + R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}} - \epsilon_n \\ & \overset{(a)}{\leq} \frac{1}{n} \Big[I\big(M_{\mathcal{Y}}^{(p)}; Y^n, M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}\big) + I\big(M, M_{\mathcal{Z}}; Z^n, M_{\mathcal{Y}}^{(c)}\big) \Big] \\ & \overset{(b)}{=} I\big(M_{\mathcal{Y}}^{(p)}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Q\big) \\ & \quad + I\big(M, M_{\mathcal{Z}}; Z_Q \big| M_{\mathcal{Y}}^{(c)}, Z^{Q-1}, Q\big) \\ & \overset{(c)}{=} I\big(M_{\mathcal{Y}}^{(p)}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q\big) \\ & \quad - I\big(Z^{Q-1}; Y_Q \big| M, M_{\mathcal{Y}}, M_{\mathcal{Z}}, Y_{Q+1}^n, Q\big) \\ & \quad + I\big(Z^{Q-1}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Q\big) \\ & \quad + I\big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q; Z_Q\big) \\ & \quad - I\big(M_{\mathcal{Y}}^{(c)}, Z^{Q-1}, Q; Z_Q\big) \end{aligned} \tag{69}$$

$$
\begin{aligned}
&- I\big(Y_{Q+1}^n; Z_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Z^{Q-1}, Q\big) && (70)\\
&\overset{(d)}{\leq} I\big(M_{\mathcal{Y}}^{(p)}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q\big)\\
&\quad + I\big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q; Z_Q\big) && (71)\\
&\overset{(e)}{\leq} H\big(Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q\big)\\
&\quad + I\big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q; Z_Q\big) && (72)\\
&\overset{(f)}{=} H(Y|U) + I(U;Z), && (73)
\end{aligned}
$$

where (a) follows from Fano's inequality, (b) is due to the chain-rule and the independence of M, $M_{\mathcal{Y}}^{(p)}$, $M_{\mathcal{Y}}^{(c)}$, and $M_{\mathcal{Z}}$, (c) is due to the chain-rule, (d) holds because of Csiszár's sum-identity and since mutual information is nonnegative, (e) is true because conditional entropy is nonnegative, and (f) is due to (62). Similarly, we have

$$
\begin{aligned}
&R + R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} - \epsilon_n\\
&\overset{(a)}{\leq} \frac{1}{n}\Big[I\big(M, M_{\mathcal{Y}}^{(c)}; Y^n, M_{\mathcal{Z}}^{(c)}\big) + I\big(M_{\mathcal{Y}}^{(p)}; Y^n, M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}\big)\\
&\quad + I\big(M_{\mathcal{Z}}^{(p)}; Z^n, M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}\big)\Big] && (74)\\
&\overset{(b)}{=} I\big(M, M_{\mathcal{Y}}^{(c)}; Y_Q \big| M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Q\big)\\
&\quad + I\big(M_{\mathcal{Y}}^{(p)}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Q\big)\\
&\quad + I\big(M_{\mathcal{Z}}^{(p)}; Z_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Z^{Q-1}, Q\big) && (75)\\
&\overset{(c)}{=} I\big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Z^{Q-1}, Q; Y_Q\big)\\
&\quad - I\big(M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Q; Y_Q\big)\\
&\quad - I\big(Z^{Q-1}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Q\big)\\
&\quad + I\big(M_{\mathcal{Y}}^{(p)}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q\big)\\
&\quad - I\big(Z^{Q-1}; Y_Q \big| M, M_{\mathcal{Y}}, M_{\mathcal{Z}}, Y_{Q+1}^n, Q\big)\\
&\quad + I\big(Z^{Q-1}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Q\big)\\
&\quad + I\big(M_{\mathcal{Z}}^{(p)}; Z_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Z^{Q-1}, Q\big)\\
&\quad - I\big(Y_{Q+1}^n; Z_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Z^{Q-1}, Q\big)\\
&\quad + I\big(Y_{Q+1}^n; Z_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Z^{Q-1}, Q\big) && (76)\\
&\overset{(d)}{\leq} I\big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Z^{Q-1}, Q; Y_Q\big)\\
&\quad + I\big(M_{\mathcal{Y}}^{(p)}; Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q\big)\\
&\quad + I\big(M_{\mathcal{Z}}^{(p)}; Z_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Z^{Q-1}, Q\big) && (77)
\end{aligned}
$$

$$
\begin{aligned}
&\overset{(e)}{\leq} I\big(M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Z^{Q-1}, Q; Y_Q\big) \\
&\quad + H\big(Y_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}, Y_{Q+1}^n, Z^{Q-1}, Q\big) \\
&\quad + I\big(M_{\mathcal{Z}}^{(p)}; Z_Q \big| M, M_{\mathcal{Y}}^{(c)}, M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Z^{Q-1}, Q\big) \qquad (78) \\
&\overset{(f)}{=} I(V;Y) + H(Y|U) + I(U;Z|V), \qquad (79)
\end{aligned}
$$

where (a) follows from Fano's inequality, (b) holds because of the chain-rule and since M, $M_{\mathcal{Y}}^{(p)}$, $M_{\mathcal{Y}}^{(c)}$, $M_{\mathcal{Z}}^{(p)}$, and $M_{\mathcal{Z}}^{(c)}$ are independent, (c) is due to the chain-rule, (d) is obtained by using Csiszár's sum-identity twice and by using that mutual information is nonnegative, (e) holds since conditional entropy is nonnegative, and (f) is due to (62). To conclude, observe that

$$
\begin{aligned}
&2R + R_{\mathcal{Y}} + R_{\mathcal{Z}} - \epsilon_n \\
&\quad \overset{(a)}{\leq} \frac{1}{n} I\big(M, M_{\mathcal{Y}}^{(c)}; Y^n, M_{\mathcal{Z}}^{(c)}\big) + H(Y|U) + I(U;Z) \qquad (80) \\
&\quad \overset{(b)}{\leq} I\big(M, M_{\mathcal{Y}}^{(c)}; Y_Q \big| M_{\mathcal{Z}}^{(c)}, Y_{Q+1}^n, Q\big) + H(Y|U) + I(U;Z) \\
&\quad \overset{(c)}{\leq} I(V;Y) + H(Y|U) + I(U;Z), \qquad (81)
\end{aligned}
$$

where (a) follows from Fano's inequality and (73), (b) is due to the chain-rule and the independence of M, $M_{\mathcal{Y}}^{(c)}$, and $M_{\mathcal{Z}}^{(c)}$, and (c) holds since conditioning cannot increase entropy.

So far, we have shown that the set of rate-tuples satisfying (1) for some PMF of the form (2) is an outer bound on the capacity region of the SD-BC with P-MSI. To conclude, it remains to establish that we can w.l.g. restrict X to be a function of (Y, U). To this end, note that by the functional representation lemma (Lemma 1), there exists a $\hat{U}$ of finite support and independent of (Y, U) such that X is a function of $(Y, U, \hat{U})$. Moreover, no upper bound in (1) decreases if we replace U by the pair $(U, \hat{U})$: each upper bound in (1) is either unaffected or increases. Hence, we can w.l.g. restrict X to be a function of (Y, U).

B A Proof of Corollary 4

For $U = X$ and $V = Y$ the constraints in (1) and (6) are equivalent. Hence, (6) is an inner bound on the capacity region. We next argue that (6) is also an outer bound on the capacity region. To this end, fix any PMF of the form (2) and where X is a function of (Y, U). By (2),

$$
I(U;Z) \leq I(X;Z). \qquad (82)
$$

Moreover,

$$
\begin{aligned}
&I(V;Y) + H(Y|U) + I(U;Z|V) \\
&\quad \overset{(a)}{=} H(Y|U) + I(U;Y,Z) - I(U;Y|V) \\
&\qquad - I(U;Z|Y) + I(U;Z|V) && (83)\\
&\quad \overset{(b)}{=} H(Y|U) + I(U;Y,Z) - I(U;Y|V) \\
&\qquad - I(X;Z|Y) + I(U;Z|V) && (84)\\
&\quad \overset{(c)}{=} H(Y|U) + I(U;Y,Z) - I(U;Y,Z|V) \\
&\qquad - I(V;Z|Y) + I(U;Z|V) && (85)\\
&\quad \overset{(d)}{\leq} H(Y|U) + I(U;Y,Z) && (86)\\
&\quad \overset{(e)}{=} H(Y) + I(U;Z|Y) && (87)\\
&\quad \overset{(f)}{=} I(X;Y,Z), && (88)
\end{aligned}
$$

where (a) follows from the chain-rule and the fact that Y, U, and V form a Markov chain in that order, (b) is true because X is a function of (Y,U) and because Z, X, and U form a Markov chain in that order, (c) holds since

$$
\begin{aligned}
I(X;Z|Y) - I(U;Z|Y,V) &= I(X;Z|Y) - I(X;Z|Y,V) && (89)\\
&= I(V;Z|Y), && (90)
\end{aligned}
$$

where we used that X is a function of (Y,U), (d) is true because mutual information is nonnegative and since conditioning cannot increase entropy, (e) follows from the chain-rule, and (f) is true because Y is a function of X, because X is a function of (Y,U), and because of the chain-rule. From (1a), (1b) and (82), as well as (1c) and (88) we conclude that (6) is also an outer bound on the capacity region.

C Analysis of Example 2

In this section, we show that if we do not bin the cloud-center, then the rate-tuple $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) = (0, p, 1-p, 0)$ is achievable only for $p \in \{0,1\}$. Recall that every rate-tuple that is achievable without binning the cloud-center satisfies (1) and (3) for some PMF of the form (2), and where one can w.l.g. restrict X to be a function of (Y,U). Now fix some PMF of the form (2) and where X is a function of (Y,U). As we argue below, (1c) implies that for $p > 0$ at least one of the following holds

$$
R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} < H(X) \leq 1 \quad \text{or} \quad I(U;X|V) = 0. \tag{91}
$$

Moreover, (3b) implies that

$$R_{\mathcal{Z}}^{(p)} \leq I(U;Z|V) \tag{92}$$
$$= I(U;Z,S|V) \tag{93}$$
$$= I(U;Z|S,V) \tag{94}$$
$$= (1-p)I(U;X|V), \tag{95}$$

where we used that S is a function of Z and independent of (V,U). On account of (91) and (95), we find that for $p>0$ at least one of the following holds

$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} < H(X) \leq 1 \quad \text{or} \quad R_{\mathcal{Z}}^{(p)} = 0. \tag{96}$$

We thus conclude that if $R_{\mathcal{Z}}^{(p)}$ is strictly positive, then $R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)}$ must be strictly smaller than 1 whenever $p>0$. In particular, this implies that if we do not bin the cloud-center, then the rate-tuple $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) = (0, p, 1-p, 0)$ is achievable only for $p \in \{0,1\}$.

To conclude, it remains to show that (1c) implies (91) for any given $p>0$. Indeed, note that every rate-tuple $(0, R_{\mathcal{Y}}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfies

$$R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} \overset{(a)}{\leq} I(V;Y) + H(Y|U) + I(U;Z|V) \tag{97}$$
$$\overset{(b)}{\leq} I(X;Y,Z) = H(X), \tag{98}$$

where (a) is (1c), and where (b) follows from (88) in the proof of Corollary 4. Moreover, equality holds only if

$$I(U;Y|Z,V) = I(V;Z|Y) = 0, \tag{99}$$

because otherwise the inequality (d) in the derivation of (88) is strict. Note that

$$I(U;Y|Z,V) = I(U;X|Z,V) \tag{100}$$
$$= (1-p)I(U;X|X,V) + pI(U;X|Z=?,V) \tag{101}$$
$$= pI(U;X|V). \tag{102}$$

For $p>0$ this proves our claim (91).

D Analysis of Example 3

In this section, we show that in the absence of MSI, i.e., when $R_{\mathcal{Y}}^{(p)} = R_{\mathcal{Y}}$ and $R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$, we can only achieve the maximum sum-rate (22) if

$$R \leq I(V;Y) \leq I(U;Y). \tag{103}$$

In the absence of MSI, it follows from Theorem 2 that the capacity region is the set of rate-tuples $(R, R_{\mathcal{Y}}, R_{\mathcal{Z}})$ satisfying

$$R + R_{\mathcal{Y}} \leq H(Y) \tag{104a}$$

$$R + R_{\mathcal{Z}} \leq I(U;Z) \tag{104b}$$

$$R + R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq I(V;Y) + H(Y|U) + I(U;Z|V) \tag{104c}$$

$$R + R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq H(Y|U) + I(U;Z) \tag{104d}$$

$$2R + R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq I(V;Y) + H(Y|U) + I(U;Z) \tag{104e}$$

for some PMF of the form (2). Because of (104e), we can only achieve the maximum sum-rate (22) if (103) holds.

E A Proof of Proposition 12

Assume $R_{\mathcal{Y}}^{(c)} > 0$; otherwise the statement is obvious. We analyze the probability of error of the described coding scheme with rate-limited feedback. Define the error events

$$E_0: \quad \text{Receiver } \mathcal{Y} \text{ reconstructs the wrong outputs } Z^{\alpha n}, \tag{105}$$

$$E_1: \quad \text{Decoding error for messages sent in Phase 1,} \tag{106}$$

$$E_2: \quad \text{Decoding error for messages (including compression index) sent in Phase 2.} \tag{107}$$

The probability of error of our scheme satisfies

$$\mathbb{P}[\text{error}] \leq \mathbb{P}[E_0 \cup E_1 \cup E_2] \tag{108}$$

$$= \mathbb{P}[E_2] + \mathbb{P}[E_0|E_2^c] + \mathbb{P}[E_1|E_0^c]\,. \tag{109}$$

Note that

$$\mathbb{P}[E_1|E_0^c] \to 0 \quad (n \to \infty) \tag{110}$$

because $(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Y}}^{(c)}, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}^{(\text{enh})}$. Also, since event E_2^c implies that receiver $\mathcal{Y}$ recovered the correct compression index and since a compression index of rate $H(Z|Y)$ is enough to recover $Z^{\alpha n}$ given the side-information $Y^{\alpha n}$,

$$\mathbb{P}[E_0|E_2^c] \to 0 \quad (n \to \infty). \tag{111}$$

Finally,

$$\mathbb{P}[E_2] \to 0 \quad (n \to \infty), \tag{112}$$

because

$$\left(R_{\mathcal{Y}}^{(p)}, \hat{R}_{\mathcal{Y}}^{(c)} + \frac{\alpha}{1-\alpha} H(Z|Y), R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}\right) \in \mathscr{C}_{\text{P-MSI}}, \tag{113}$$

where we used (33) and that $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}$. The "rate" $\frac{\alpha}{1-\alpha} H(Z|Y)$ accounts for the $\alpha n H(Z|Y)$ compression bits from Phase 1 that have to be sent during the $(1-\alpha)n$ channel uses of Phase 2.

F A Proof of Propositions 13 and 14

Before proving Propositions 13 and 14, we first sketch the proof of Proposition 13. A first step is to note that, under the conditions stated in Proposition 13, it is possible to identify a rate-triple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ that satisfies the following:

1. $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)})$ lies on the boundary of the no-feedback capacity region $\mathscr{C}$ of the SD-BC without MSI and in the interior of the capacity region $\mathscr{C}_{\text{enh}}$ of the enhanced BC without MSI.

2. There exists a strictly-positive rate $R_{\mathcal{Y}}^{(c)}$ such that $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ is contained in $\mathscr{C}_{\text{P-MSI}}$.

As we argue next, the above conditions guarantee that, irrespective of $R_{\text{FB}} > 0$, the feedback coding scheme of Section 4.1 achieves a rate-tuple outside the no-feedback capacity region $\mathscr{C}$. (Recall from Corollary 6 that $\mathscr{C}$ is also the capacity of the SD-BC with P-MSI at receiver $\mathcal{Y}$ only.) Indeed, Condition 1 guarantees that in Phase 1 of the feedback code, when the encoder codes for the enhanced BC, message information can be sent at rates $(\tilde{R}_{\mathcal{Y}}, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)})$ satisfying

$$\tilde{R}_{\mathcal{Y}} > R_{\mathcal{Y}}^{(p)}, \quad \tilde{R}_{\mathcal{Z}}^{(p)} > R_{\mathcal{Z}}^{(p)}, \quad \tilde{R}_{\mathcal{Z}}^{(c)} > R_{\mathcal{Z}}^{(c)}. \tag{114}$$

The deterministic receiver $\mathcal{Y}$ needs resolution information to decode its phase-1 message. Condition 2 ensures that in Phase 2 of the feedback code the encoder can transmit the required resolution information at rate $R_{\mathcal{Y}}^{(c)}$ and simultaneously send fresh message information at rates $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$. Specifically, Proposition 12 implies that for α sufficiently small the rate-tuple

$$\alpha(\tilde{R}_{\mathcal{Y}}, 0, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}) + (1-\alpha)(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \tag{115}$$

is achievable. Since $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)})$ lies on the boundary of the no-feedback capacity region $\mathscr{C}$, we conclude from (114) that the rate-tuple (115) is not achievable without feedback, and that feedback thus increases capacity.

The proof of Proposition 14 is similar: the only difference is that we set $R_{\mathcal{Z}}^{(c)} = \tilde{R}_{\mathcal{Z}}^{(c)} = 0$.

We next provide the details of the proof of Propositions 13 and 14. We shall need the following lemma:

Lemma 18. *Fix a rate-tuple* $(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}$. *If* $R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$ *(or, equivalently,* $R_{\mathcal{Z}}^{(c)} > 0$*), and if for some PMF* $p(u, x, y, z)$ *of the form* (12) *we have*

$$R_{\mathcal{Y}}^{(p)} < H(Y) \tag{116a}$$

$$R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)} \leq I(U; Z) \tag{116b}$$

$$R_{\mathcal{Y}}^{(p)} + R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)} \leq H(Y|U) + I(U; Z) \tag{116c}$$

$$I(U; Y) > 0, \tag{116d}$$

then there exists a positive rate $R_{\mathcal{Y}}^{(c)} > 0$ *satisfying*

$$(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}. \tag{117}$$

If $R_{\mathcal{Z}}^{(p)} = R_{\mathcal{Z}}$ *(or, equivalently,* $R_{\mathcal{Z}}^{(c)} = 0$*), and if for some PMF* $p(v, u, x, y, z)$ *of the form* (2) *we have* (116) *and*

$$I(V; Y) - I(V; Z) > 0, \tag{118}$$

then there exists a positive rate $R_{\mathcal{Y}}^{(c)} > 0$ *satisfying* (117).

Proof of Lemma 18. Suppose that (116) holds for some PMF $p(u, x, y, z)$ of the form (12). The rate-tuple $\big(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}\big)$ is in $\mathscr{C}_{\text{P-MSI}}$ if it satisfies (1) for some conditional PMF $p(v|u)$ and for

$$p(v, u, x, y, z) = p(v|u)p(u, x, y, z). \tag{119}$$

By inspection of (1) and (116), we find that this holds if

$$\begin{aligned}
R_{\mathcal{Y}}^{(c)} &\leq H(Y) - R_{\mathcal{Y}}^{(p)} \\
R_{\mathcal{Y}}^{(c)} &\leq H(Y|U) + I(U; Z) + I(V; Y) - I(V; Z) - R_{\mathcal{Y}}^{(p)} - R_{\mathcal{Z}}^{(p)} \\
R_{\mathcal{Y}}^{(c)} &\leq H(Y|U) + I(U; Z) + I(V; Y) - R_{\mathcal{Y}}^{(p)} - R_{\mathcal{Z}}^{(p)} - R_{\mathcal{Z}}^{(c)}.
\end{aligned}$$

On account of (116), there exists some $R_{\mathcal{Y}}^{(c)} > 0$ that satisfies the above inequalities if

$$I(V; Z) - I(V; Y) < R_{\mathcal{Z}}^{(c)}, \quad I(V; Y) > 0. \tag{121}$$

If $R_{\mathcal{Z}}^{(c)} = 0$, then (121) and (118) are equivalent. Suppose now that $R_{\mathcal{Z}}^{(c)} > 0$. Fix $\varepsilon \in (0,1)$, let $S \sim \mathrm{Ber}(1-\varepsilon)$ be independent of (U, X, Y, Z), and choose

$$V = \begin{cases} U, & S = 0, \\ ?, & S = 1. \end{cases} \tag{122}$$

Note that (121) holds for ε sufficiently small because

$$I(V;Z) = \varepsilon I(U;Z) \leq \varepsilon \log |\mathcal{Z}|, \quad 0 < I(V;Y) = \varepsilon I(U;Y).$$

This proves the claim for $R_{\mathcal{Z}}^{(c)} > 0$. □

Note that Lemma 18 does not ask for much when $R_{\mathcal{Z}}^{(p)} < R_{\mathcal{Z}}$: Since $(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}$, there is a PMF $p(u,x,y,z)$ of the form (12) for which (116b) and (116c) are satisfied and (116a) holds with non-strict inequality. Hence, all we are asking for is that $R_{\mathcal{Y}} < H(Y)$ and $I(U;Y) > 0$. Roughly speaking, we ask that the transmission of $M_{\mathcal{Z}}$ interfere the transmission of $M_{\mathcal{Y}}$.

Proof of Proposition 13 and 14. Assume that one can find a rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ that satisfies the following three conditions:[3]

$$R_{\mathcal{Y}}^{(c)} > 0 \tag{123a}$$

$$(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}} \tag{123b}$$

$$(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)} + R_{\mathcal{Z}}^{(c)}) \in \big(\partial\mathscr{C} \cap (\mathscr{C}_{\text{enh}} \setminus \partial\mathscr{C}_{\text{enh}})\big). \tag{123c}$$

Look now for rates

$$\tilde{R}_{\mathcal{Y}}^{(p)} > R_{\mathcal{Y}}^{(p)}, \quad \tilde{R}_{\mathcal{Z}}^{(p)} > R_{\mathcal{Z}}^{(p)}, \quad \tilde{R}_{\mathcal{Z}}^{(c)} > R_{\mathcal{Z}}^{(c)} \tag{124}$$

such that $(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(p)} + \tilde{R}_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{enh}}$. By Assumption (123c) such rates must exist.

Proposition 12 and Assumptions (123a) and (123b) guarantee that, irrespective of $R_{\text{FB}} > 0$, there exists an $\alpha \in (0,1)$ such that the rate-tuple

$$\alpha(\tilde{R}_{\mathcal{Y}}^{(p)}, 0, \tilde{R}_{\mathcal{Z}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(c)}) + (1-\alpha)(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \tag{125}$$

is in the feedback capacity region of the SD-BC with P-MSI at receiver $\mathcal{Y}$. Note that (10), (123c), and (124) imply that the rate-tuple in (125) is not in $\mathscr{C}_{\text{P-MSI}}$ (see also Footnote 4). To conclude the proof of Proposition 13, it thus suffices to show that, under

[3] In particular, (123c) implies that $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \partial\mathscr{C}_{\text{P-MSI}}^{(R_{\mathcal{Y}}^{(c)}=0)}$, where $\mathscr{C}_{\text{P-MSI}}^{(R_{\mathcal{Y}}^{(c)}=0)} \subset (\mathbb{R}_0^+)^3$ denotes the set of rate-triples $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying $(R_{\mathcal{Y}}^{(p)}, 0, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)}) \in \mathscr{C}_{\text{P-MSI}}$ (cf. (10)).

the conditions in Proposition 13, it is possible to find a rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, R_{\mathcal{Z}}^{(c)})$ satisfying (123). But this follows from the first part of Lemma 18.

The proof of Proposition 14 is similar: Assume that one can find a rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, 0)$ that satisfies the following three conditions:

$$R_{\mathcal{Y}}^{(c)} > 0 \tag{126a}$$

$$(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, 0) \in \mathscr{C}_{\text{P-MSI}} \tag{126b}$$

$$(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}) \in \big(\partial\mathscr{C} \cap (\mathscr{C}_{\text{enh}} \setminus \partial\mathscr{C}_{\text{enh}})\big). \tag{126c}$$

Look now for rates

$$\tilde{R}_{\mathcal{Y}}^{(p)} > R_{\mathcal{Y}}^{(p)}, \quad \tilde{R}_{\mathcal{Z}}^{(p)} > R_{\mathcal{Z}}^{(p)} \tag{127}$$

such that $(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(p)}) \in \mathscr{C}_{\text{enh}}$. By Assumption (126c) such rates must exist.

Proposition 12 and Assumptions (126a) and (126b) guarantee that, irrespective of $R_{\text{FB}} > 0$, there exists an $\alpha \in (0, 1)$ such that the rate-pair

$$\alpha(\tilde{R}_{\mathcal{Y}}^{(p)}, \tilde{R}_{\mathcal{Z}}^{(p)}) + (1 - \alpha)(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Z}}^{(p)}) \tag{128}$$

is in the feedback capacity region of the SD-BC with P-MSI at receiver $\mathcal{Y}$. Note that (126c) and (127) imply that the rate-tuple in (128) is not in $\mathscr{C}$. To conclude the proof of Proposition 13, it thus suffices to show that, under the conditions in Proposition 13, it is possible to find a rate-tuple $(R_{\mathcal{Y}}^{(p)}, R_{\mathcal{Y}}^{(c)}, R_{\mathcal{Z}}^{(p)}, 0)$ satisfying (126). But this follows from the second part of Lemma 18. □

G Analysis of Example 4

The proof of Example 4 hinges on Propositions 13 and 14, which state sufficient conditions for feedback to increase the capacity of the SD-BC with and without P-MSI at receiver $\mathcal{Y}$. It can be roughly outlined as follows: For the SD-BC of Example 4, we first parametrize $\partial\mathscr{C} \cap \big\{R_{\mathcal{Z}} \in [R_{\mathcal{Z}}^*, \bar{p}]\big\}$ by $R_{\mathcal{Z}}$, where $R_{\mathcal{Z}}^*$ denotes the maximum rate $R_{\mathcal{Z}}$ for which we can achieve the sum-rate capacity of the SD-BC without MSI, and where $\bar{p}$ is the maximum rate $R_{\mathcal{Z}}$ that is achievable on the SD-BC without MSI. Then, we show that for the boundary point with $R_{\mathcal{Z}} = R_{\mathcal{Z}}^*$ we can satisfy all conditions in Proposition 13, and that for some boundary point with $R_{\mathcal{Z}} \in (R_{\mathcal{Z}}^*, \bar{p})$ we can satisfy all conditions in Proposition 14. Hence, Propositions 13 and 14 imply that feedback increases the sum-rate capacity of the SD-BC with P-MSI at the deterministic receiver $\mathcal{Y}$

and the capacity region of the SD-BC without MSI.

Recall that the no-feedback capacity region $\mathscr{C}$ of the SD-BC without MSI is the set of rate-tuples satisfying (4) for some PMF of the form (5), and where we can restrict X to be a function of (Y, U). For the SD-BC of Example 4, the fact that X is a function of (Y, U) implies that we can partition the support $\mathcal{U}$ of U into two disjoint sets $\mathcal{U}_0$ and $\mathcal{U}_1$, where

$$p_{X_2,Y|U}(1,1|u) = 0,\ \forall\, u \in \mathcal{U}_0 \quad \text{and} \quad p_{X_2,Y|U}(0,1|u) = 0,\ \forall\, u \in \mathcal{U}_1. \tag{129}$$

For each $u \in \mathcal{U}_0$ define

$$p_0(u) \triangleq p_{X_2,Y|U}(0,0|u) \tag{130a}$$
$$p_1(u) \triangleq p_{X_2,Y|U}(0,1|u) \tag{130b}$$
$$p_2(u) \triangleq p_{X_2,Y|U}(1,2|u), \tag{130c}$$

and note that $p_0(u) + p_1(u) + p_2(u) = 1$. As we argue next, we can w.l.g. assume that to each $u \in \mathcal{U}_0$ there corresponds a $u' \in \mathcal{U}_1$ satisfying $p_U(u) = p_U(u')$ and

$$p_0(u) = p_{X_2,Y|U}(0,0|u) = p_{X_2,Y|U}(1,2|u') \tag{131a}$$
$$p_1(u) = p_{X_2,Y|U}(0,1|u) = p_{X_2,Y|U}(1,1|u') \tag{131b}$$
$$p_2(u) = p_{X_2,Y|U}(1,2|u) = p_{X_2,Y|U}(0,0|u'), \tag{131c}$$

This allows us to simplify (4b) and (4c) to

$$R_{\mathcal{Z}} \leq \bar{p}\Big[1 - 2\sum_{u\in\mathcal{U}_0} p(u) h_{\mathrm{b}}(p_2(u))\Big], \tag{132a}$$
$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq \bar{p} + 2\sum_{u\in\mathcal{U}_0} p(u)\Big[p h_{\mathrm{b}}(p_2(u)) + (1-p_2(u)) h_{\mathrm{b}}\Big(\frac{p_0(u)}{p_0(u)+p_1(u)}\Big)\Big]. \tag{132b}$$

Indeed, note that for all $u \in \mathcal{U}_0$ and $u' \in \mathcal{U}_1$ satisfying (131) we have $H(X_2|U=u) = H(X_2|U=u')$ and $H(Y|U=u) = H(Y|U=u')$. The claim now follows by symmetry, since $I(U;Z) = \bar{p} I(U;X_2)$, and since entropy is concave.

With (132) at hand, we determine the sum-rate capacity. First, note that the RHS of (132b) is maximum only if $p_0(u) = p_1(u) = \big(1 - p_2(u)\big)/2$, and in this case (132b) simplifies to

$$R_{\mathcal{Y}} + R_{\mathcal{Z}} \leq \bar{p} + 2\sum_{u\in\mathcal{U}_0} p(u)\big[p h_b(p_2(u)) + (1-p_2(u))\big]. \tag{133}$$

Moreover, the function $p_2(u) \mapsto ph_b(p_2(u)) + \big(1 - p_2(u)\big)$ is strictly concave in $p_2(u)$, and hence we readily find that the RHS of (133) is maximum if and only if for all $u \in \mathcal{U}_0$ we have

$$p_2(u) = \frac{1}{1 + 2^{1/p}}. \tag{134}$$

Hence, we find that the sum-rate capacity is obtained by evaluating the RHS of (133) for the choice (134), and that every PMF $p(u, x, y, z)$ of the form (5) that achieves the maximum sum-rate satisfies the following

$$0 < I(U; Y) < I(U; Z) \tag{135a}$$

$$R_{\mathcal{Z}} \leq I(U; Z) = \bar{p}\left[1 - h_b\left(\frac{1}{1 + 2^{1/p}}\right)\right] = R^*_{\mathcal{Z}}, \tag{135b}$$

where $R^*_{\mathcal{Z}}$ is the maximum rate $R_{\mathcal{Z}}$ for which we can achieve the sum-rate capacity of the SD-BC without MSI.

We next parametrize $\partial\mathscr{C} \cap \big\{R_{\mathcal{Z}} \in [R^*_{\mathcal{Z}}, \bar{p}]\big\}$ by $R_{\mathcal{Z}}$, i.e., for each $R_{\mathcal{Z}} \in [R^*_{\mathcal{Z}}, \bar{p}]$ we determine

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = \max\big\{R_{\mathcal{Y}} \geq 0 \colon (R_{\mathcal{Y}}, R_{\mathcal{Z}}) \in \mathscr{C}\big\}. \tag{136}$$

We start by showing that $R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}}$ is strictly decreasing in $R_{\mathcal{Z}}$. Indeed, because $\mathscr{C}$ is convex we have for all $\epsilon \in (0, R_{\mathcal{Z}} - R^*_{\mathcal{Z}}]$, for $\alpha = \epsilon/(R_{\mathcal{Z}} - R^*_{\mathcal{Z}})$, and for $\bar{\alpha} = 1 - \alpha$

$$\begin{aligned} &R_{\mathcal{Y}}(R_{\mathcal{Z}} - \epsilon) + R_{\mathcal{Z}} - \epsilon \\ &\quad \geq \bar{\alpha}\big(R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}}\big) + \alpha\big(R_{\mathcal{Y}}(R^*_{\mathcal{Z}}) + R^*_{\mathcal{Z}}\big) && (137) \\ &\quad > R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}}, && (138) \end{aligned}$$

where the last inequality holds because $R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}}$ is maximum if $R_{\mathcal{Z}} = R^*_{\mathcal{Z}}$ and strictly smaller than its maximum if $R_{\mathcal{Z}} > R^*_{\mathcal{Z}}$. As we argue next, each pair $(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}})$ satisfies

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = H(Y|U) \tag{139a}$$

$$R_{\mathcal{Z}} = I(U; Z) \tag{139b}$$

for some PMF $p(u, x, y, z)$ of the form (5). Note that the claim holds for $R_{\mathcal{Z}} = R^*_{\mathcal{Z}}$, and hence we assume that $R^*_{\mathcal{Z}} < R_{\mathcal{Z}}$. Since $(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}}) \in \partial\mathscr{C}$, Corollary 3 implies that for some PMF $p(u, x, y, z)$ of the form (5) we have

$$R_{\mathcal{Z}} \leq I(U; Z) \tag{140}$$

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = \min\{H(Y|U) + I(U; Z) - R_{\mathcal{Z}}, H(Y)\}. \tag{141}$$

For contradiction, assume that $R_{\mathcal{Z}} < I(U;Z)$. Then,

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}} \overset{(a)}{\geq} \bar{\alpha}\big(H(Y|U) + I(U;Z)\big) + \alpha\big(R_{\mathcal{Y}}(R^*_{\mathcal{Z}}) + R^*_{\mathcal{Z}}\big) \tag{142}$$

$$\overset{(b)}{>} R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}}, \tag{143}$$

where (a) holds for $\alpha = (I(U;Z) - R_{\mathcal{Z}})/(I(U;Z) - R^*_{\mathcal{Z}})$ and $\bar{\alpha} = 1 - \alpha$ because $R_{\mathcal{Z}} = \bar{\alpha} I(U;Z) + \alpha R^*_{\mathcal{Z}}$ and because the capacity region is convex, and (b) is true because $R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}} \leq H(Y|U) + I(U;Z)$, since $\alpha > 0$, and because $R^*_{\mathcal{Z}} < R_{\mathcal{Z}}$ and $R_{\mathcal{Y}}(R_{\mathcal{Z}}) + R_{\mathcal{Z}}$ is strictly decreasing in $R_{\mathcal{Z}}$. This is cleary a contradiction, and hence the claim follows.

From (132), (133), and (139) we obtain that each pair $(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}})$, $R_{\mathcal{Z}} \in [R^*_{\mathcal{Z}}, \bar{p}]$ is of the form

$$R_{\mathcal{Z}} = \bar{p}\Big[1 - 2\sum_{u \in \mathcal{U}_0} p(u) h_{\mathrm{b}}(p_2(u))\Big], \tag{144a}$$

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = \bar{p} + 2\sum_{u \in \mathcal{U}_0} p(u)\big[p h_{\mathrm{b}}(p_2(u)) + 1 - p_2(u)\big] - R_{\mathcal{Z}}. \tag{144b}$$

This can also be written as

$$R_{\mathcal{Z}} = \sum_{u \in \mathcal{U}_0} 2p(u)\bar{p}[1 - h_{\mathrm{b}}(p_2(u))] \tag{145a}$$

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = \sum_{u \in \mathcal{U}_0} 2p(u)[h_{\mathrm{b}}(p_2(u)) + 1 - p_2(u)], \tag{145b}$$

where we used that

$$\sum_{u \in \mathcal{U}_0} 2p(u) = 1. \tag{146}$$

Note that if $p_2(u) > 1/2$, then replacing $p_2(u)$ by $1 - p_2(u)$ in (145) does not affect the value of $R_{\mathcal{Z}}$ but increases that of $R_{\mathcal{Y}}(R_{\mathcal{Z}})$. Hence, for each $u \in \mathcal{U}_0$ satisfying that $p(u) > 0$ we have $p_2(u) \in [0, 1/2]$, and hence we can w.l.g. assume that $p_2(u) \in [0, 1/2]$ holds for all $u \in \mathcal{U}_0$. Since $h_{\mathrm{b}}(\cdot)$ is invertible on $[0, 1/2]$ and $\bar{p}$ is a constant, we can thus write

$$R_{\mathcal{Z}} = \sum_{u \in \mathcal{U}_0} 2p(u) R^{(u)}_{\mathcal{Z}} \tag{147a}$$

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = \sum_{u \in \mathcal{U}_0} 2p(u)\left[2 - \frac{1}{\bar{p}} R^{(u)}_{\mathcal{Z}} - h_{\mathrm{b}}^{-1}\left(1 - \frac{R^{(u)}_{\mathcal{Z}}}{\bar{p}}\right)\right], \tag{147b}$$

where $R^{(u)}_{\mathcal{Z}} = \bar{p}[1 - h_{\mathrm{b}}(p_2(u))]$. Note that $R^{(u)}_{\mathcal{Z}}$ can assume any value in $[0, \bar{p}]$ depending on $p_2(u) \in [0, 1/2]$. Since $h_{\mathrm{b}}(\cdot)$ is strictly increasing and strictly concave on $[0, 1/2]$,

its inverse $h_{\mathrm{b}}^{-1}(\cdot)$ is strictly increasing and strictly convex on $[0,1]$. In particular, this implies that the mapping

$$x \mapsto 2 - \frac{1}{\bar{p}}x - h_b^{-1}\left(1 - \frac{x}{\bar{p}}\right) \tag{148}$$

is strictly concave on $[0,1]$. Recalling that $R_{\mathcal{Y}}(R_{\mathcal{Z}})$ is the maximum rate $R_{\mathcal{Y}}$ that is achievable for a given $R_{\mathcal{Z}}$, we thus obtain from Jensen's inequality that for each $u \in \mathcal{U}_0$ satisfying that $p(u) > 0$ we must have $R_{\mathcal{Z}}^{(u)} = R_{\mathcal{Z}}$. This eventually allows us to parametrize $\partial\mathscr{C} \cap \left\{R_{\mathcal{Z}} \in [R_{\mathcal{Z}}^*, \bar{p}]\right\}$ by $R_{\mathcal{Z}}$:

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = 2 - \frac{1}{\bar{p}}R_{\mathcal{Z}} - h_{\mathrm{b}}^{-1}\left(1 - \frac{R_{\mathcal{Z}}}{\bar{p}}\right),\ R_{\mathcal{Z}} \in [R_{\mathcal{Z}}^*, \bar{p}]. \tag{149}$$

With (149) at hand, we next argue that for $p > 1/2$ there exists a rate-pair $(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}})$ with $R_{\mathcal{Z}} < \bar{p}$ such that (4) and (38) of Proposition 14 hold for some PMF $p(u,x,y,z)$ of the form (5) and for $V = U$. Because $(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}}) \in \partial\mathscr{C}$ and since for $V = U$ the PMF $p(v,u,x,y,z)$ is of the form (2), this is almost enough to conclude from Proposition 14 that feedback increases the capacity of the SD-BC without MSI: once we have established the claim, all that remains to show is that the identified rate-pair satisfies

$$(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}}) \in \mathscr{C}_{\mathrm{enh}} \setminus \partial\mathscr{C}_{\mathrm{enh}}. \tag{150}$$

To establish the claim, let $\mathcal{U}_0 = \{u\}$. In particular, this implies that $\mathcal{U}_1 = \{u'\}$, where (131) holds and $p(u) = p(u') = 1/2$. For every $R_{\mathcal{Z}} \in [R_{\mathcal{Z}}^*, \bar{p}]$ set $p_2(u) = h_{\mathrm{b}}^{-1}(1 - R_{\mathcal{Z}}/\bar{p})$ and

$$\frac{1-p_2(u)}{2} = p_{X_2,Y|U}(0,0|u) = p_{X_2,Y|U}(1,2|u') \tag{151a}$$

$$\frac{1-p_2(u)}{2} = p_{X_2,Y|U}(0,1|u) = p_{X_2,Y|U}(1,1|u') \tag{151b}$$

$$p_2(u) = p_{X_2,Y|U}(1,2|u) = p_{X_2,Y|U}(0,0|u'). \tag{151c}$$

For this choice we have

$$R_{\mathcal{Y}}(R_{\mathcal{Z}}) = H(Y|U) = h_{\mathrm{b}}(p_2(u)) + 1 - p_2(u) \tag{152a}$$

$$R_{\mathcal{Z}} = I(U;Z) = \bar{p}(1 - h_{\mathrm{b}}(p_2(u))), \tag{152b}$$

Moreover, it holds that

$$H(Y) = h_{\mathrm{b}}\left(\frac{1-p_2(u)}{2}\right) + \frac{1+p_2(u)}{2}, \tag{153}$$

and from (152a) we thus obtain

$$I(U;Y) = H(Y) - H(Y|U) \tag{154}$$

$$= h_{\mathrm{b}}\left(\frac{1-p_2(u)}{2}\right) - h_{\mathrm{b}}(p_2(u)) + \frac{3p_2(u)-1}{2}. \tag{155}$$

Note that for all $p_2(u) \in [0, 1/(1+2^{1/p})]$ we have

$$R_{\mathcal{Y}} < H(Y). \tag{156}$$

Moreover, if $p > 1/2$, then we obtain for $p_2(u) = 0$ that

$$I(U;Z) = \bar{p} < \frac{1}{2} = I(U;Y). \tag{157}$$

Since $I(U;Y)$ and $I(U;Z)$ are continuous in $p_2(u)$, this implies that there exists an open interval $(0,\kappa) \subset \big(0, 1/(1+2^{1/p})\big)$ such that

$$I(U;Y) > I(U;Z),\ p_2(u) \in (0,\kappa). \tag{158}$$

Indeed, this implies our claim: for $p > 1/2$ there exists a rate-pair $(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}})$ with $R_{\mathcal{Z}} < \bar{p}$ such that (4) and (38) of Proposition 14 hold for some PMF $p(u,x,y,z)$ of the form (5) and for $V = U$.

As we argue next, to conclude the analysis of Example 4 it now suffices to show that on the enhanced BC $\big(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}}\big) \in \mathscr{C}_{\mathrm{enh}} \setminus \partial\mathscr{C}_{\mathrm{enh}}$ holds for all $R_{\mathcal{Z}} \in [R^*_{\mathcal{Z}}, \bar{p})$. Indeed, we already showed that this is enough to conclude from Proposition 14 that feedback increases the capacity region of the SD-BC without MSI. Moreover, it is also enough to conclude from Proposition 13 that feedback increases the sum-rate capacity of the SD-BC with P-MSI at the deterministic receiver $\mathcal{Y}$. To see this, note that $\big(R_{\mathcal{Y}}(R^*_{\mathcal{Z}}), R^*_{\mathcal{Z}}\big) \in \partial\mathscr{C}$ and that (11) and (36) hold for some PMF of the form (12), where (36) follows form (135a) and (139).

To conclude, it now remains to show that on the enhanced BC $\big(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}}\big) \in \mathscr{C}_{\mathrm{enh}} \setminus \partial\mathscr{C}_{\mathrm{enh}}$ holds for all $R_{\mathcal{Z}} \in [R^*_{\mathcal{Z}}, \bar{p})$. Let $U \sim \mathrm{Ber}(1/2)$ be a binary random variable and set

$$\frac{1-p_2(u)}{2} = p_{X_2,Y|U}(0,0|0) = p_{X_2,Y|U}(1,2|1) \tag{159a}$$

$$\frac{1-p_2(u)}{2} = p_{X_2,Y|U}(0,1|0) = p_{X_2,Y|U}(1,1|1) \tag{159b}$$

$$\epsilon p_2(u) = p_{X_2,Y|U}(1,1|0) = p_{X_2,Y|U}(0,1|1) \tag{159c}$$

$$(1-\epsilon)p_2(u) = p_{X_2,Y|U}(1,2|0) = p_{X_2,Y|U}(0,0|1), \tag{159d}$$

where ϵ and $p_2(u)$ take value in the set $[0,1]$. If $\epsilon = 0$ and $p_2(u) \in \big[0, 1/(1+2^{1/p})\big]$, then, under the above PMF $p(u, x_2, y)$, (11) and (31) both evaluate to (152). Note that

$$I(U;Z) = \bar{p}\big[1 - h_b(p_2(u))\big] \tag{160}$$

does not depend on ϵ but $I(X;Y,Z|U)$ does. Define

$$\begin{aligned} f\big(\epsilon, p_2(u)\big) &\triangleq I(X;Y,Z|U) && (161)\\ &= \bar{p}H(X_1,X_2|U) + pH(Y|U) \\ &= \bar{p}\big[h_b(p_2(u)) + 1 - p_2(u) + p_2(u)h_b(\epsilon)\big] \\ &\quad - p\Bigg[\frac{1-p_2(u)}{2}\log\bigg(\frac{1-p_2(u)}{2}\bigg) \\ &\quad + \bigg(\frac{1-(1-2\epsilon)p_2(u)}{2}\bigg)\log\bigg(\frac{1-(1-2\epsilon)p_2(u)}{2}\bigg) \\ &\quad + (1-\epsilon)p_2(u)\log\big((1-\epsilon)p_2(u)\big)\Bigg]. && (162) \end{aligned}$$

Note that for all $p_2(u) \in \big(0, 1/(1+2^{1/p})\big]$ we have

$$\begin{aligned} \frac{\partial f\big(\epsilon, p_2(u)\big)}{\partial \epsilon} &= \bar{p}p_2(u)\log\bigg(\frac{1-\epsilon}{\epsilon}\bigg) + pp_2(u)\log\bigg(\frac{2\bar{\epsilon}p_2(u)}{1-(1-2\epsilon)p_2(u)}\bigg) \\ &\overset{\epsilon\downarrow 0}{\to} \infty. \end{aligned}$$

In particular, this implies that there exists an $\epsilon > 0$ so that $f\big(\epsilon, p_2(u)\big) > f\big(0, p_2(u)\big)$. Since the capacity region of the enhanced SD-BC contains every rate-pair $(\tilde{R}_{\mathcal{Y}}, \tilde{R}_{\mathcal{Z}})$ that satisfies for some $\epsilon > 0$

$$\tilde{R}_{\mathcal{Y}} = I(X;Y,Z|U) \tag{163a}$$

$$\tilde{R}_{\mathcal{Z}} = I(U;Z), \tag{163b}$$

this proves our claim that for all $R_{\mathcal{Z}} \in [R^*_{\mathcal{Z}}, \bar{p})$ we have $\big(R_{\mathcal{Y}}(R_{\mathcal{Z}}), R_{\mathcal{Z}}\big) \in \mathscr{C}_{\text{enh}} \setminus \partial\mathscr{C}_{\text{enh}}$.

Remark 3. *To prove that feedback can increase the capacity region of the SD-BC without MSI, we used Proposition 14 with the choice $V = U$. This raises the question if choosing $V \neq U$ can help. It turns out that this is indeed the case. To see this, extend the SD-BC of Example 4 by a parallel channel of capacity larger than* $\log 3$*, and assume that receiver $\mathcal{Z}$ additionally observes the output of that parallel channel. (E.g., assume that, in addition to Z, receiver $\mathcal{Z}$ noiselessly observes an input X_3, which takes value in a set of size* 4*.) For the constructed SD-BC, it is easy to see that a rate-tuple satisfying*

(11) *for some PMF of the form* (12) *is a boundary point of the capacity region only if the auxiliary random variable* U *comprises a capacity-achieving input to the parallel channel. (E.g., if receiver* $\mathcal{Z}$ *observes the pair* (Z, X_3)*, then* X_3 *must have a uniform prior and be deterministic given* U*.) But this implies that* $I(U;Y) \leq \log 3 < I(U;Z)$*. Hence, we cannot invoke Proposition 14 with the choice* $V = U$ *to show that feedback can increase capacity. However the following choice of* V *does the trick: let* V *be the random variable that is obtained by discarding the input to the parallel channel from* U*.*

H A Proof of Theorem 16

To prove the theorem, we must show that every rate-tuple in the feedback capacity region of the SD-BC with F-MSI at the stochastic receiver $\mathcal{Z}$ satisfies (6) for some PMF of the form (7). Let $Q \sim \text{Unif}[1:n]$ be independent of $(M_{\mathcal{Y}}, M_{\mathcal{Z}})$, and define $(X, Y, Z) \triangleq (X_Q, Y_Q, Z_Q)$.

The rate of message $M_{\mathcal{Y}}$ satisfies

$$R_{\mathcal{Y}} - \epsilon_n \overset{(a)}{\leq} \frac{1}{n} I\big(M_{\mathcal{Y}}; Y^n, M_{\mathcal{Z}}^{(c)}\big) \tag{164}$$

$$\overset{(b)}{\leq} H\big(Y_Q \big| M_{\mathcal{Z}}^{(c)}, Y^{Q-1}, Q\big) \tag{165}$$

$$\overset{(c)}{\leq} H(Y), \tag{166}$$

where (a) follows from Fano's inequality, (b) is true because of the chain-rule, because $M_{\mathcal{Y}}$ and $M_{\mathcal{Z}}^{(c)}$ are independent, and because conditional entropy is nonnegative, and (c) holds since conditioning cannot increase entropy. The rate of message $M_{\mathcal{Z}}$ satisfies

$$R_{\mathcal{Z}} - \epsilon_n \overset{(a)}{\leq} \frac{1}{n} I(M_{\mathcal{Z}}; Z^n, M_{\mathcal{Y}}) \tag{167}$$

$$\overset{(b)}{\leq} I(M_{\mathcal{Z}}; Z_Q | M_{\mathcal{Y}}, Z^{Q-1}, Q) \tag{168}$$

$$\overset{(c)}{\leq} I(X; Z), \tag{169}$$

where (a) follows form Fano's inequality, (b) is true because of the chain-rule and because $M_{\mathcal{Y}}$ and $M_{\mathcal{Z}}$ are independent, and (c) holds since conditioning cannot increase entropy and since Z, X, and $(M_{\mathcal{Y}}, M_{\mathcal{Z}}, Z^{Q-1}, Q)$ form a Markov chain in that order.

To establish the sum-rate constraint, note that if the probability of a decoding error is small, then $M_{\mathcal{Y}}$ is with high probability computable from $(Y^n, M_{\mathcal{Z}}^{(c)})$ and $M_{\mathcal{Z}}^{(p)}$ from $(Z^n, M_{\mathcal{Y}})$ and thus from $(Y^n, Z^n, M_{\mathcal{Z}}^{(c)})$. Hence, the probability of a decoding error is

small only if $(M_{\mathcal{Y}}, M_{\mathcal{Z}}^{(p)})$ is with high probability computable from $(Y^n, Z^n, M_{\mathcal{Z}}^{(c)})$, and thus we conclude that

$$
\begin{aligned}
&R_{\mathcal{Y}} + R_{\mathcal{Z}}^{(p)} - \epsilon_n \\
&\quad \overset{(a)}{\leq} \frac{1}{n} I\big(M_{\mathcal{Y}}, M_{\mathcal{Z}}^{(p)}; Y^n, Z^n, M_{\mathcal{Z}}^{(c)}\big) && (170)\\
&\quad \overset{(b)}{=} I\big(M_{\mathcal{Y}}, M_{\mathcal{Z}}^{(p)}; Y_Q, Z_Q \big| M_{\mathcal{Z}}^{(c)}, Y^{Q-1}, Z^{Q-1}, Q\big) && (171)\\
&\quad \overset{(c)}{\leq} I(X; Y, Z), && (172)
\end{aligned}
$$

where (a) follows from Fano's inequality, (b) is due to the chain-rule and the independence of $\big(M_{\mathcal{Y}}, M_{\mathcal{Z}}^{(p)}\big)$ and $M_{\mathcal{Z}}^{(c)}$, and (c) holds since conditioning cannot increase entropy and since (Y, Z), X, and $(M_{\mathcal{Y}}, M_{\mathcal{Z}}, Y^{Q-1}, Z^{Q-1}, Q)$ form a Markov chain in that order.

I Analysis of Example 5

On account of Corollary 6, we know that the capacity region without feedback is the set of rate-tuples satisfying (11) for some PMF $p(u, x, y, z)$ of the form (12), where we can w.l.g. restrict X to be a function of (Y, U). Note that

$$
\begin{aligned}
I(U; Z) &\overset{(a)}{=} I(U; Z, S) && (173)\\
&\overset{(b)}{=} I(U; Z|S) && (174)\\
&\overset{(c)}{=} \bar{p} I(U; X) && (175)\\
&\overset{(d)}{=} \bar{p} I(U; Y) + \bar{p} I(U; X|Y) && (176)\\
&\overset{(e)}{=} \bar{p} I(U; Y) + \bar{p} H(X|Y) && (177)\\
&= \bar{p} I(U; Y) + \bar{p} \sum_{y \in \mathcal{Y}} p(y) H(X|Y = y) && (178)\\
&\overset{(f)}{\leq} \bar{p} I(U; Y) + \bar{p} \sum_{y \in \mathcal{Y}} p(y) \log|\mathcal{X}_y|, && (179)
\end{aligned}
$$

where (a) holds since S is computable from Z, (b) is due to the chain-rule and the independence of S and U, (c) follows from (39), (d) holds because Y is a function of X and because of the chain-rule, (e) is true because X is a function of (Y, U), and (f) is true because $Y = y$ implies that $X \in \mathcal{X}_y$, and because the uniform distribution maximizes entropy. Note that equality in (f) can be achieved irrespective of $H(Y)$ and

$I(U;Y)$. Since

$$\begin{aligned}&H(Y|U)+\bar{p}I(U;Y)+\bar{p}\sum_{y\in\mathcal{Y}}p(y)\log|\mathcal{X}_y|\\&\quad=H(Y)-pI(U;Y)+\bar{p}\sum_{y\in\mathcal{Y}}p(y)\log|\mathcal{X}_y|,\end{aligned}\tag{180}$$

we conclude that the capacity region without feedback is indeed the set of rate-tuples $(R_{\mathcal{Y}},0,R_{\mathcal{Z}}^{(p)},R_{\mathcal{Z}}^{(c)})$ satisfying (40) for some PMF $p(u,x,y,z)$ of the form (12).

Consider now the case with feedback. As we argue next, the feedback capacity region of any SD-BC without MSI at the stochastic receiver $\mathcal{Z}$ ($R_{\mathcal{Y}}^{(p)}=R_{\mathcal{Y}}$) is contained in the set of all rate-tuples satisfying (11a), (11b), and

$$R_{\mathcal{Y}}+R_{\mathcal{Z}}\leq I(X;Y,Z|U)+I(U;Z)\tag{181}$$

for some PMF $p(u,x,y,z)$ of the form (12), irrespective of whether or not the deterministic receiver $\mathcal{Y}$ has MSI ($R_{\mathcal{Z}}^{(p)}\in[0,R_{\mathcal{Z}}]$). Indeed, let $Q\sim\text{Unif}[1:n]$ be independent of $(M_{\mathcal{Y}},M_{\mathcal{Z}})$, and define

$$U_Q\triangleq(M_{\mathcal{Z}},Y^{Q-1},Z^{Q-1}),\ U\triangleq(U_Q,Q),\tag{182a}$$

$$X\triangleq X_Q,\ Y\triangleq Y_Q,\ Z\triangleq Z_Q.\tag{182b}$$

Note that, also in the presence of feedback, U, X, and (Y,Z) form a Markov chain in that order, i.e., their PMF is of the form (12). Using Fano's inequality, we readily obtain the claim:

$$\begin{aligned}R_{\mathcal{Y}}-\epsilon_n&\leq\frac{1}{n}I(M_{\mathcal{Y}};Y^n,M_{\mathcal{Z}})\leq H(Y),\\R_{\mathcal{Z}}-\epsilon_n&\leq\frac{1}{n}I(M_{\mathcal{Z}};Z^n)\leq I(U;Z),\\R_{\mathcal{Y}}+R_{\mathcal{Z}}-\epsilon_n&\leq\frac{1}{n}\big[I(M_{\mathcal{Y}};Y^n,Z^n,M_{\mathcal{Z}})+I(M_{\mathcal{Z}};Z^n)\big]\\&\leq I(X;Y,Z|U)+I(U;Z).\end{aligned}$$

To conclude that feedback cannot increase capacity, we finally note that for the considered SD-BC

$$I(X;Y,Z|U)+I(U;Z)$$

$$\overset{(a)}{=}H(Y|U)+I(X;Z|Y,U)+I(U;Z)\tag{183}$$

$$\overset{(b)}{=}H(Y|U)+I(X;Z,S|Y,U)+I(U;Z,S)\tag{184}$$

$$\overset{(c)}{=}H(Y|U)+I(X;Z|Y,U,S)+I(U;Z|S)\tag{185}$$

$$\stackrel{(d)}{=} H(Y|U) + \bar{p}H(X|Y,U) + \bar{p}I(U;X) \tag{186}$$

$$\stackrel{(e)}{=} H(Y|U) + \bar{p}H(X|Y,U) + \bar{p}I(U;Y) + \bar{p}I(U;X|Y) \tag{187}$$

$$= H(Y) - pI(U;Y) + \bar{p}H(X|Y,U) + \bar{p}I(U;X|Y) \tag{188}$$

$$\stackrel{(f)}{=} H(Y) - pI(U;Y) + \bar{p}H(X|Y) \tag{189}$$

$$\stackrel{(g)}{\leq} H(Y) - pI(U;Y) + \bar{p}\sum_{y\in\mathcal{Y}} p(y)\log|\mathcal{X}_y|, \tag{190}$$

where (a) holds because of the chain-rule and since Y is a function of X, (b) is true because S is computable from Z, (c) is due to the chain-rule and the independence of S and (U,X), (d) follows from (39), (e) holds because Y is a function of X and because of the chain-rule, (f) is due to the chain-rule, and (g) is true because $Y = y$ implies that $X \in \mathcal{X}_y$, and because the uniform distribution maximizes entropy.

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